

CDE2501 | Liveable Cities Seminar 1

17 January 2025

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Outline of today's seminar

- **Why CDE2501 Liveable Cities? How is it taught?**
- **What's Liveability Framework (LF)?**
- **Transport Case Study**
- **Q&A**

Why CDE2501 Liveable Cities?

Why study cities?



The world is rapidly urbanised



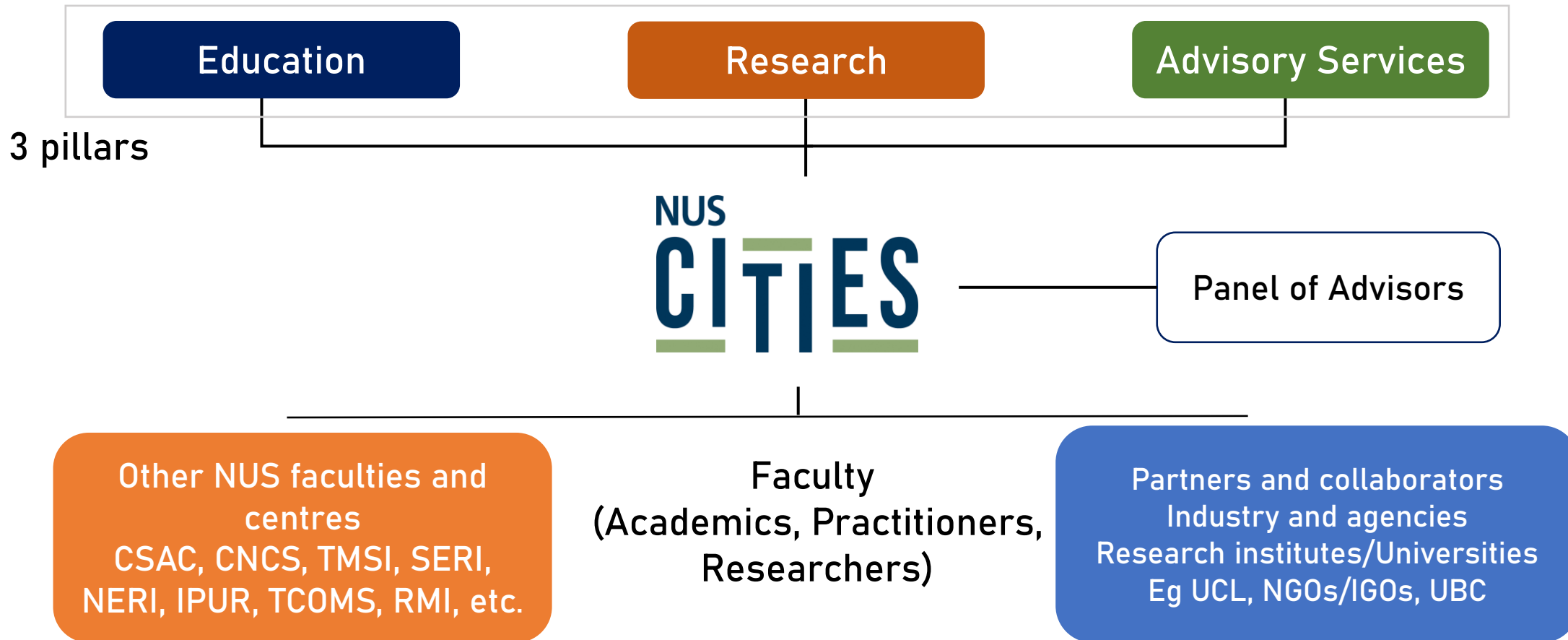
Cities are likely to determine the future sustainability of the world



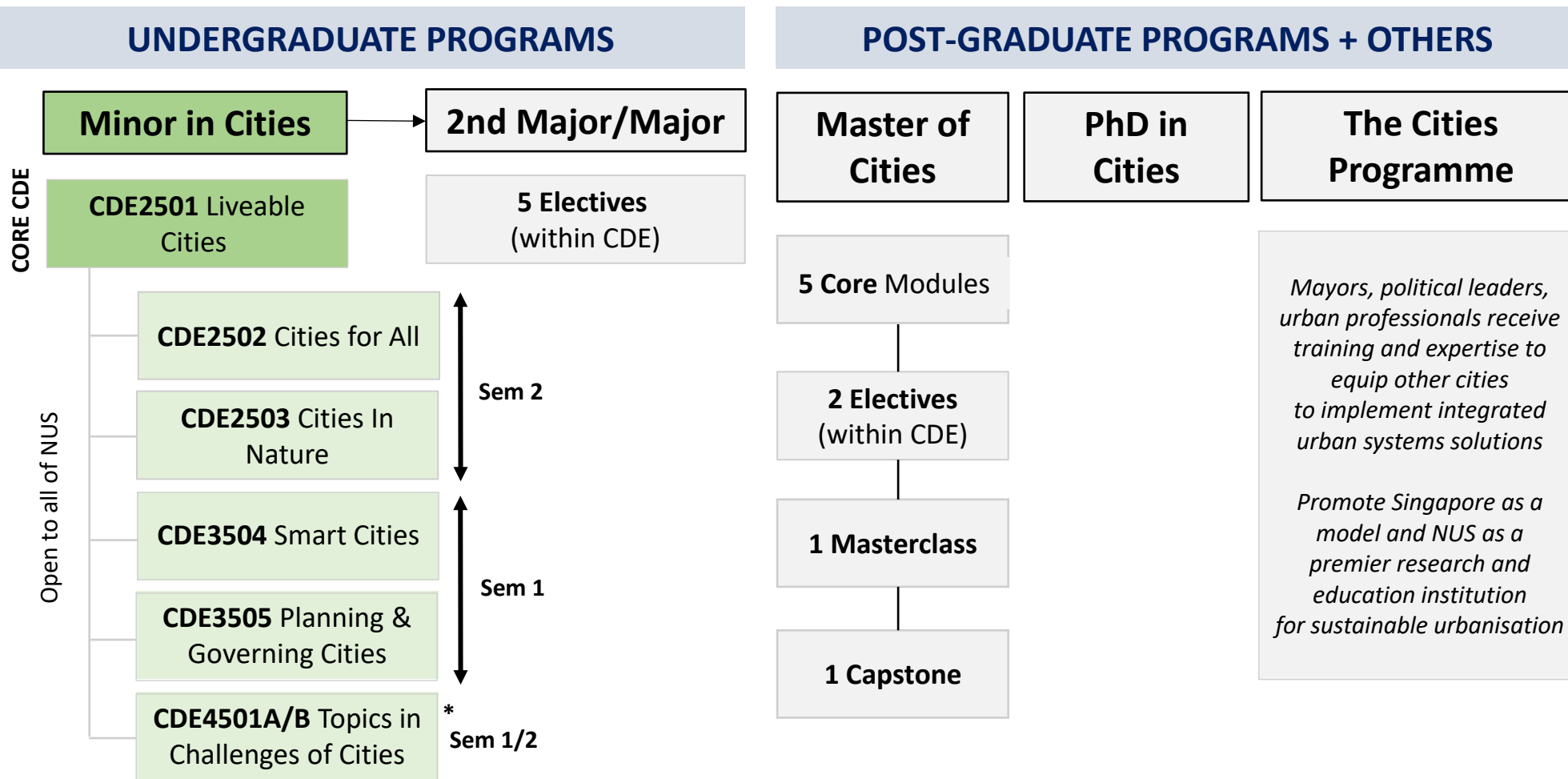
Singapore is an island city state facing existential challenges

NUS Cities advances the know-hows

A multi-disciplinary, open, and inclusive collaborative platform



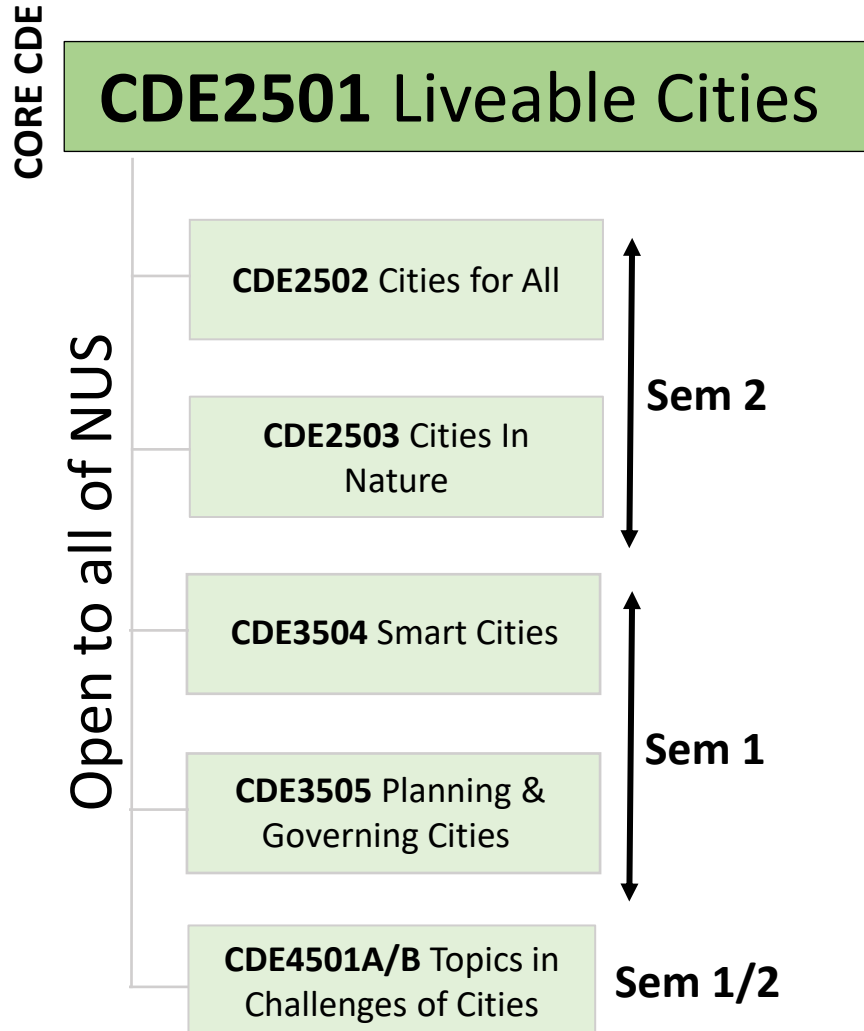
IN DEVELOPMENT



*depends on visiting professorship periods

Source: NUS Cities, 2025

Do consider pursuing a Minor in Cities



CORE COURSES (20 Units)

The **five core courses** stem from the Centre for Liveable Cities (CLC) Liveability Framework and build upon our core curriculum course **CDE2501 Liveable Cities**

CDE2502 Cities For All: Ageing, Inclusiveness, Quality of Life, hardware/software/heartware

CDE2503 Cities in Nature: Blue and Green systems, Sustainable Environment, Food Waste, Energy Resilience, Climate Adaptation

CDE3504 Smart Cities: Urban Economics, Knowledge Economy, Digital Economy, Industry, Smart Nation, ULI

CDE3505 Planning & Governance: Rule of Law, Long term Planning, Local Planning, Data Analytics, Governance, Policy Frameworks

ELECTIVES (4 Units)

Elective courses change every semester to feature visiting professor(s) and their respective fields of expertise

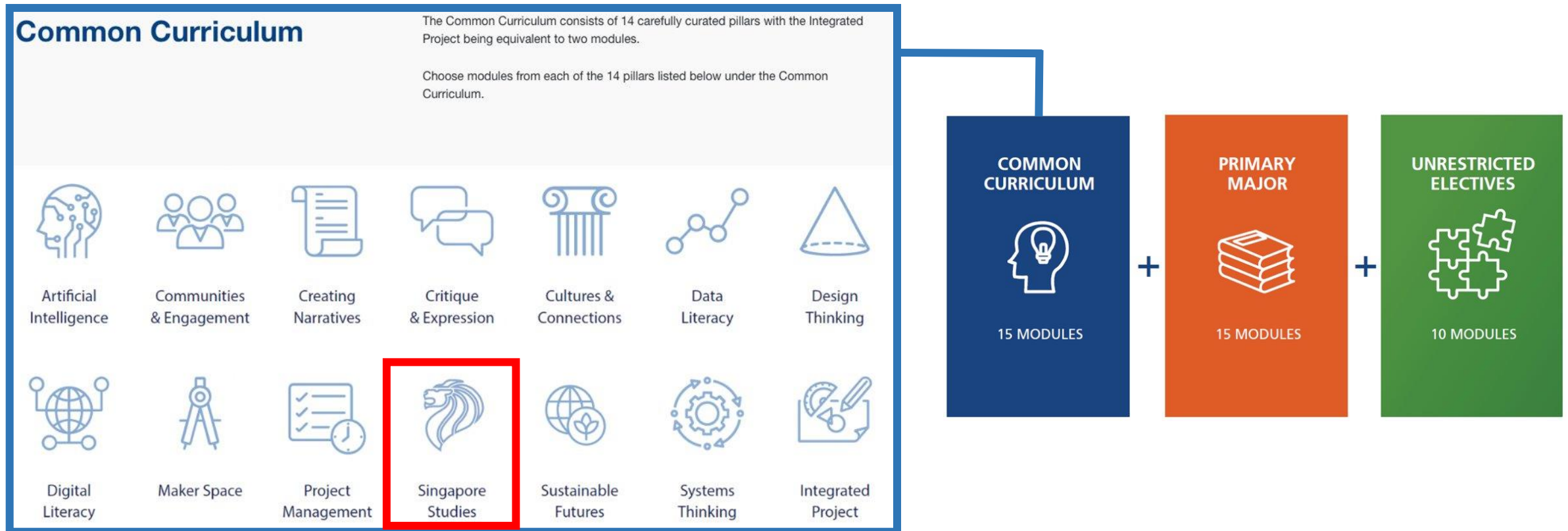
CDE4501A/B Topics in Challenges of Cities*

**Theme of CDE4501A/B will vary in each semester; and will depend on the periods of visiting professorship periods.*

Where CDE2501 Liveable Cities (4 Units) sits

For CDE students: Common Curriculum | CDE Interdisciplinary courses

For non-CDE students: Singapore Studies Elective | Unrestricted Elective Courses



Source: NUS Cities, 2025

What's the relevance to CDE disciplines?

Architecture
Biomedical Engineering
Chemical Engineering
Civil Engineering
Computer Engineering
Electrical Engineering
Engineering Science
Environmental & Sustainability Engineering
Industrial Engineering
Industrial and Systems Engineering
Infrastructure and Project Management
Landscape Architecture
Materials Science & Engineering
Mechanical Engineering
Robotics and Machine Intelligence

- **Solving complex problems requires multi-disciplinary efforts and expertise**
 - Ability to connect the dots and relate them to higher purposes or broader perspectives
 - Ability to work in teams and collaborate effectively across job boundaries
- **Prepare you to be better “T” professionals:**
 - **Horizontal ability** to gain the breadth of knowledge and to understand the “big picture”
 - **Vertical ability** to contribute your depth of domain expertise and to innovate in problem-solving
- **Proxy the reality of work for your learning before your employment**

Your views

How this course is taught

Your two take-aways from this course

Course objectives

- **Understand** how the **urban systems** approach has helped Singapore achieve its **liveability outcomes**
- **Appreciate** your **role as future professionals** to solve **urban problems**

Seminars, studios & Liveability Framework (LF)

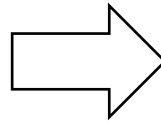
Learning urban systems through team projects

SEMINARS

- 6 Seminars, 2 hours each
- Thematic case studies
- 3 quizzes

STUDIOS

- 7 studios, 3 hours each
- Team projects
- 2 presentations

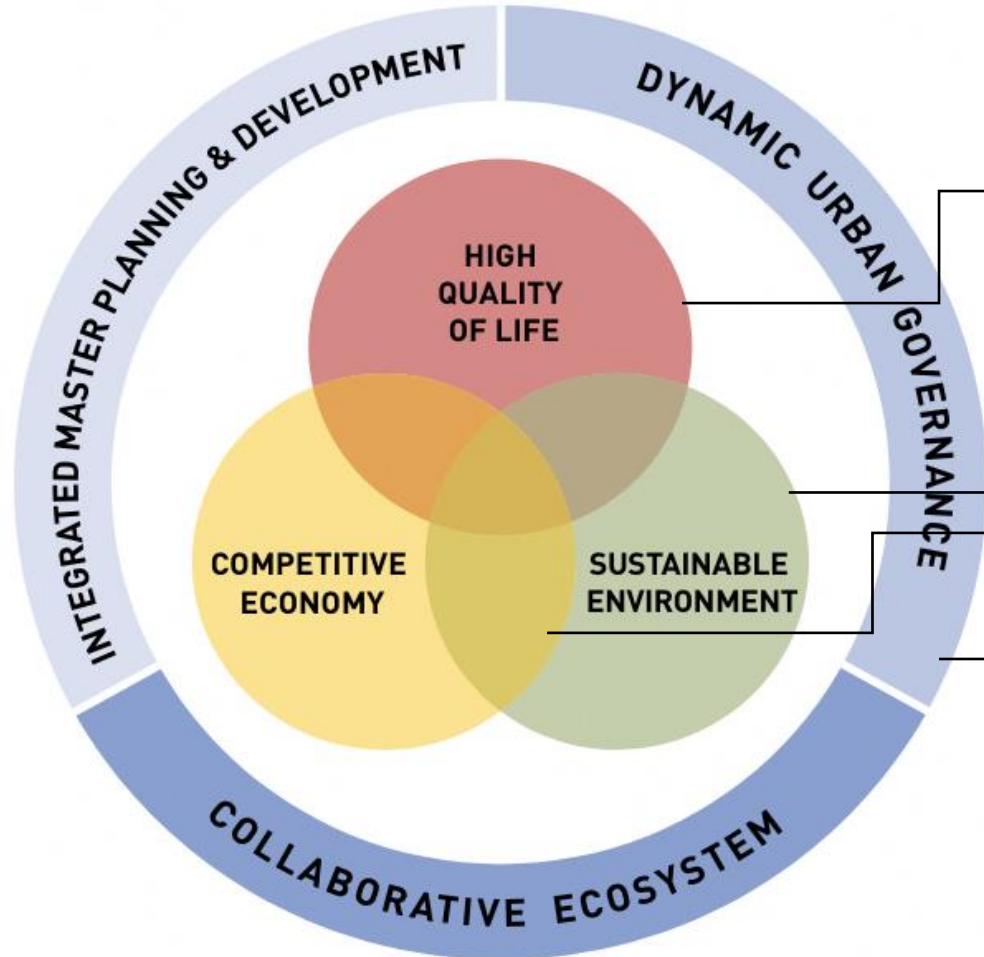


*Relate and apply
your learning to
your **team projects***

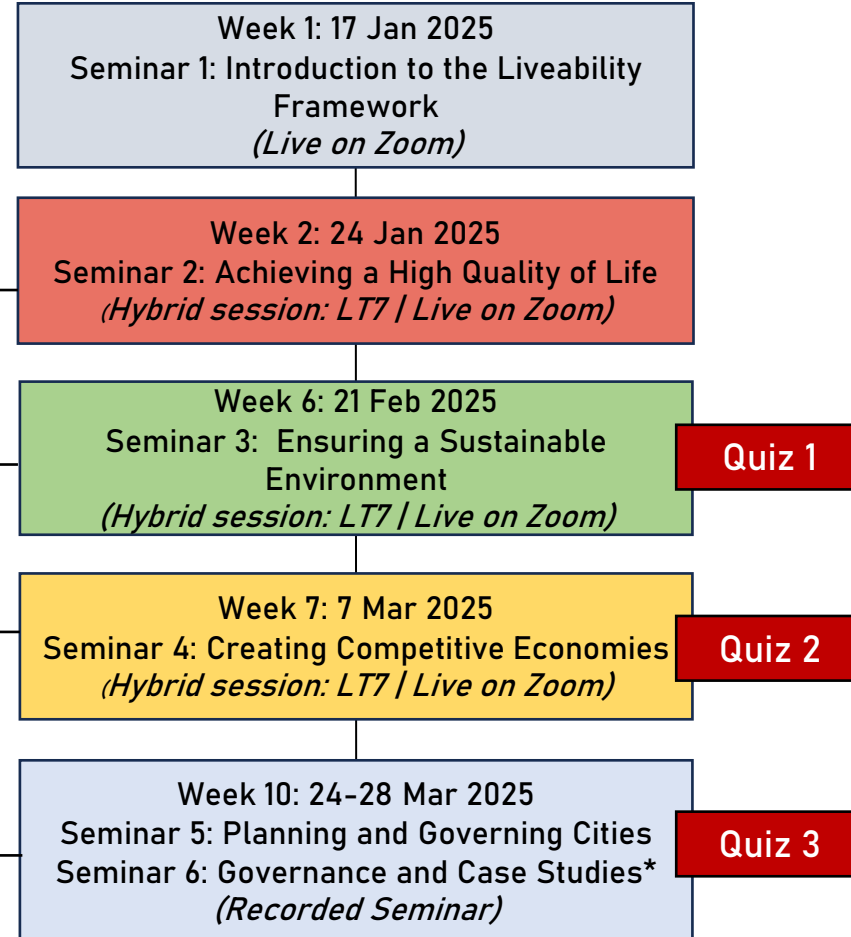
*Liveability Framework
Outcomes, Systems and Enablers*



Seminars



SEMINARS Fridays 10 AM - 12 PM



*NUS Well-Being Day on 28 Mar 2025.
Recorded Seminar will be used instead.

Learn from the Seminars

Case-sharing

- Delivered by practitioners and academics on selected case studies

Link to the Studios

- Connect content and discussions to learning outcomes and studio projects

Panel discussions

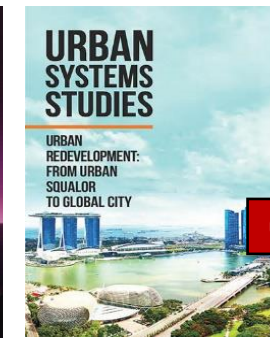
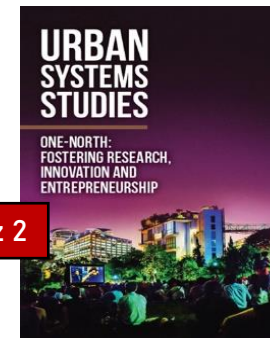
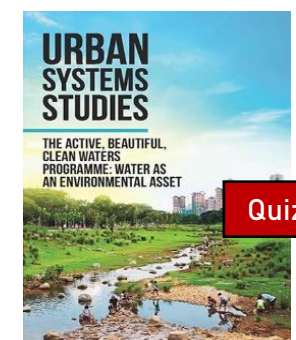
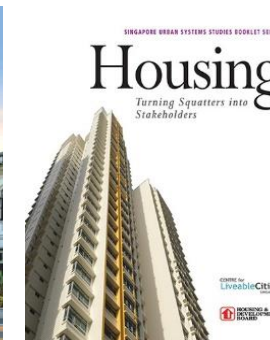
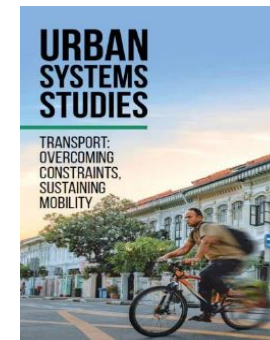
- Open Q&A

Quiz (5 MCQs)

- Held during seminar time
- Test students' knowledge of content of assigned reading for that week

Thematic Readings

Urban Systems Studies (USS)



Fridays 10:00am – 12:00pm (hybrid) in LT 7 (480 pax)
Students will be assigned to in-person seminars

Studios for you to work on team projects

CONSULTATION STUDIO



- All teams come prepared with work done before the studio and be ready for short sharing in class
- Review project work done and project ideas with SL/TA
- Mini discussion to guide you through the deliverables

PRESENTATION STUDIO



- Each team is given 10 mins (12 mins for Final Presentation) to showcase your work and pitch your projects
- Q&A sessions with peers and SL/TA for cross-learning
- One (the same) project site throughout the course



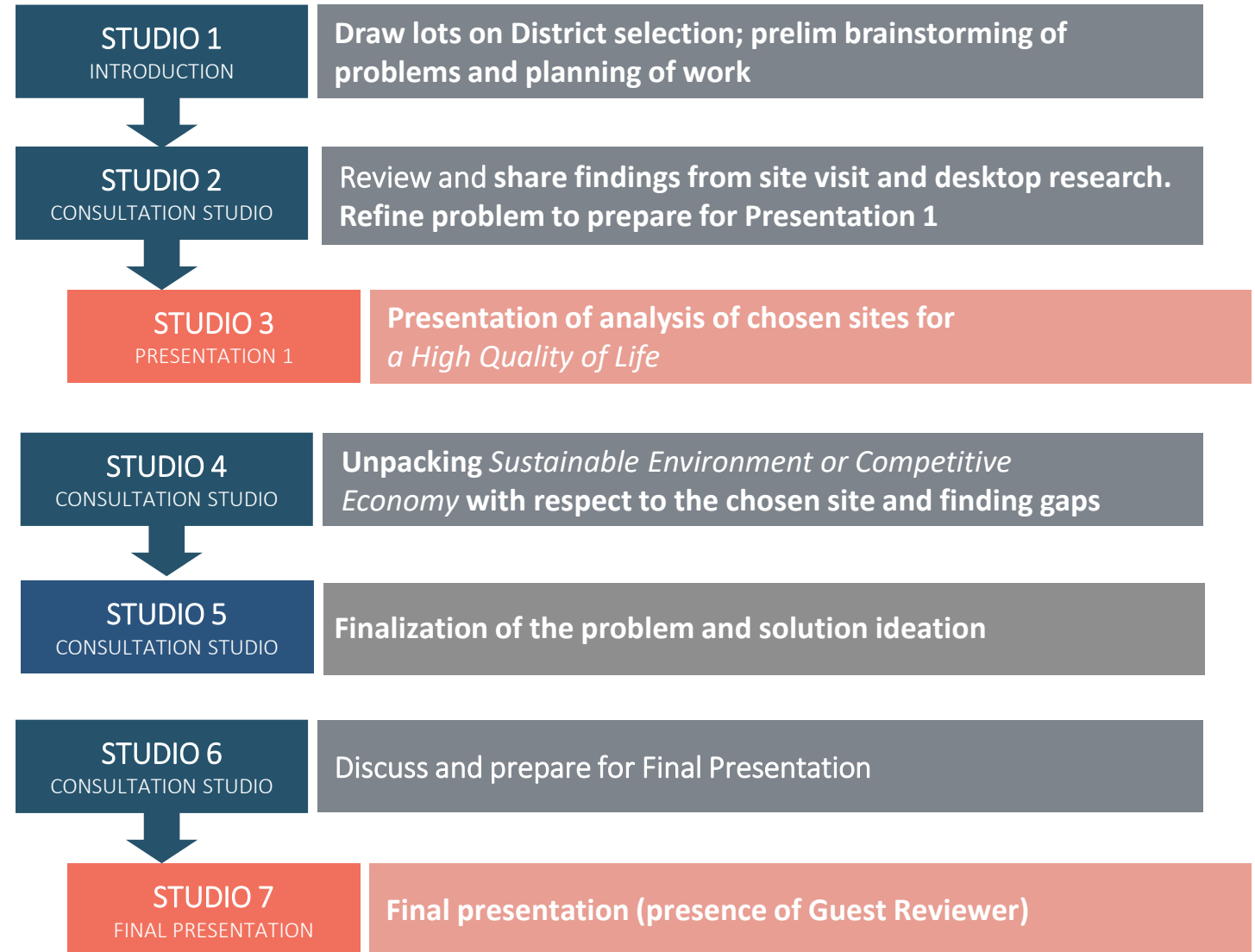
Demonstrate your:

- *Understanding of urban systems to achieve outcomes*
- *Appreciation of future roles in your team projects*

Your Studios line-up

Introductory Studio | Studio 1
Consultation Studio | Studio 2, 4, 5, 6
Presentation Studio | Studio 3, 7*

***Final Presentation (Studio 7):**
 Each team to make a pitch to
 a **guest reviewer**



Your overall Course schedule

Week	Seminar / Studio
1	[17 Jan 2025] Seminar 1: Introduction
2	[24 Jan 2025] Seminar 2: Achieving a High Quality of Life
3	[28 Jan – 31 Jan 2025] Studio 1: Introductory Studio
4	[4 Feb – 7 Feb 2025] Studio 2: Consultation Studio
5	[11 Feb – 14 Feb 2025] Studio 3: Presentation Studio
6	[21 Feb 2025] Seminar 3: Ensuring a Sustainable Environment
RECESS WEEK	

Quiz 1

Week	Seminar / Studio
7	[7 Mar 2025] Seminar 4: Creating Competitive Economies
8	[11 Mar – 14 Mar 2025] Studio 4: Consultation Studio
9	[18 Mar – 21 Mar 2025] Studio 5: Consultation Studio
10	[24-28 Mar 2025] Seminar 5: Planning and Governing Cities* Seminar 6: Governance and Case Studies* (Recorded due to clash with NUS Wellbeing Day)
11	[1 Apr - 4 Apr 2025] Studio 6: Consultation Studio
12	[8 Apr – 11 Apr 2025] Studio 7: Presentation Studio

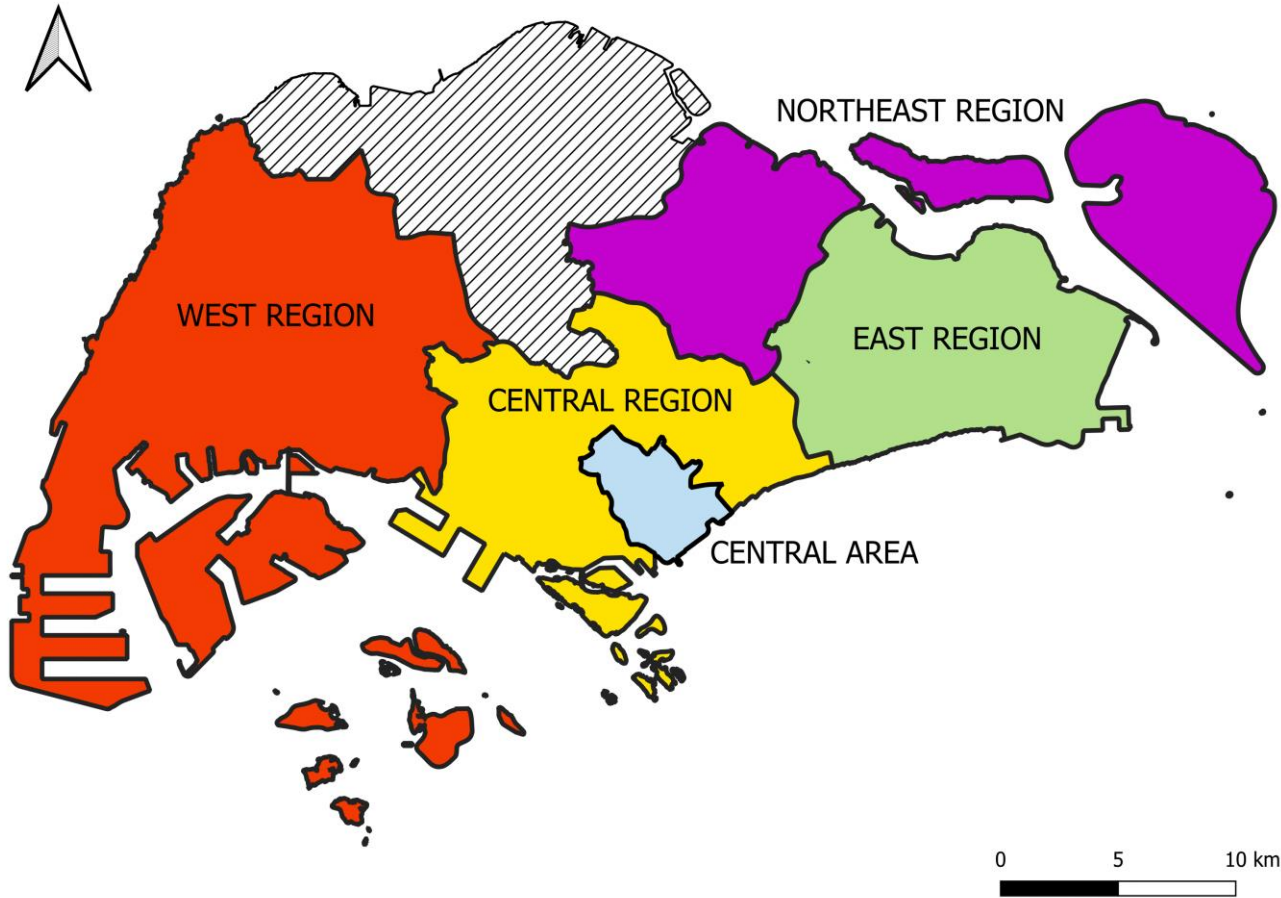
Quiz 2

Quiz 3

Source: NUS Cities, 2025

*Recorded Seminar will be uploaded to Canvas for students to learn at their own time.

What your team project is about



- Teams are formed as assigned
- Each team will draw lots (in Studio 1) to pick one of five regions* in Singapore
- Each team has a choice of two defined Sites (in each region) to select
- Each team will then work on the **same chosen Site**, and make **2 Presentations**, throughout the whole course

** Regions are the same as URS's planning regions.*

Make full use of Studio & Site Guides



A Studio Guide and 5 Site Guides are your useful resource available in Canvas

Refer to them for info about the course, Project Sites and suggested prompts* (issues about the Sites)

- Team can also choose other issues to work on

Use such info and prompts to help you work on your team projects

If you have queries, please approach your Studio Leaders (SLs) and Teaching Assistants (TAs)

Source: NUS Cities, 2025

Maximise your team endeavour

- Your team is **multi-disciplinary**
 - Reality of future work environment
- Listen to others and contribute your views to make “**the whole greater than the sum of its parts**”
- Make use of the **varied strengths of team members**
- Project-manage, “divide-and-conquer” to **maximise outputs**
- Make the best out of and **enjoy the friendship**
 - Your paths may cross in the future!



Image source: Googled stock picture, 2025

What will be assessed ?

ASSESSMENT CRITERIA	WEIGHTAGE		
Seminar Attendance	5%		
Studio Project	85%	Studio Attendance	5%
		Individual Performance	20%
		Group Presentation 1	30%
		Final Group Presentation*	30%
Quiz (x3)	10%		

*Presence of Guest Reviewer

Please refer to the Studio Guide for details on each studio's deliverables and evaluation rubrics

Individual Performance comprises

- Amount of Active Participation
- Quality of Active Participation
- Quality of Individual Presentation
- Peer Review Score

Tips:

- **Don't throw away** your attendance, performance and quiz marks.
- **Don't be a free-loader** or else you won't get the same Group Grade as your team-mates!

CDE2501 Core Team

Co-Course Leaders



**Adjunct Prof.
Looi Teik Soon**

Former Advisor and Dean,
LTA Academy & Singapore Rail Academy



**Assoc. Prof.
Wee Hiang Koon**

Founding and Principal Architect at
SKEW Collaborative
Registered Architect



Dr. Rosita Samsudin

Lecturer, NUS Cities
PhD, Architecture

CDE2501 Core Team

Team of Course Coordinators



Thomas Kent Bean
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Teaching Assistant, NUS Cities
BA (Hons) Urban Studies, Yale
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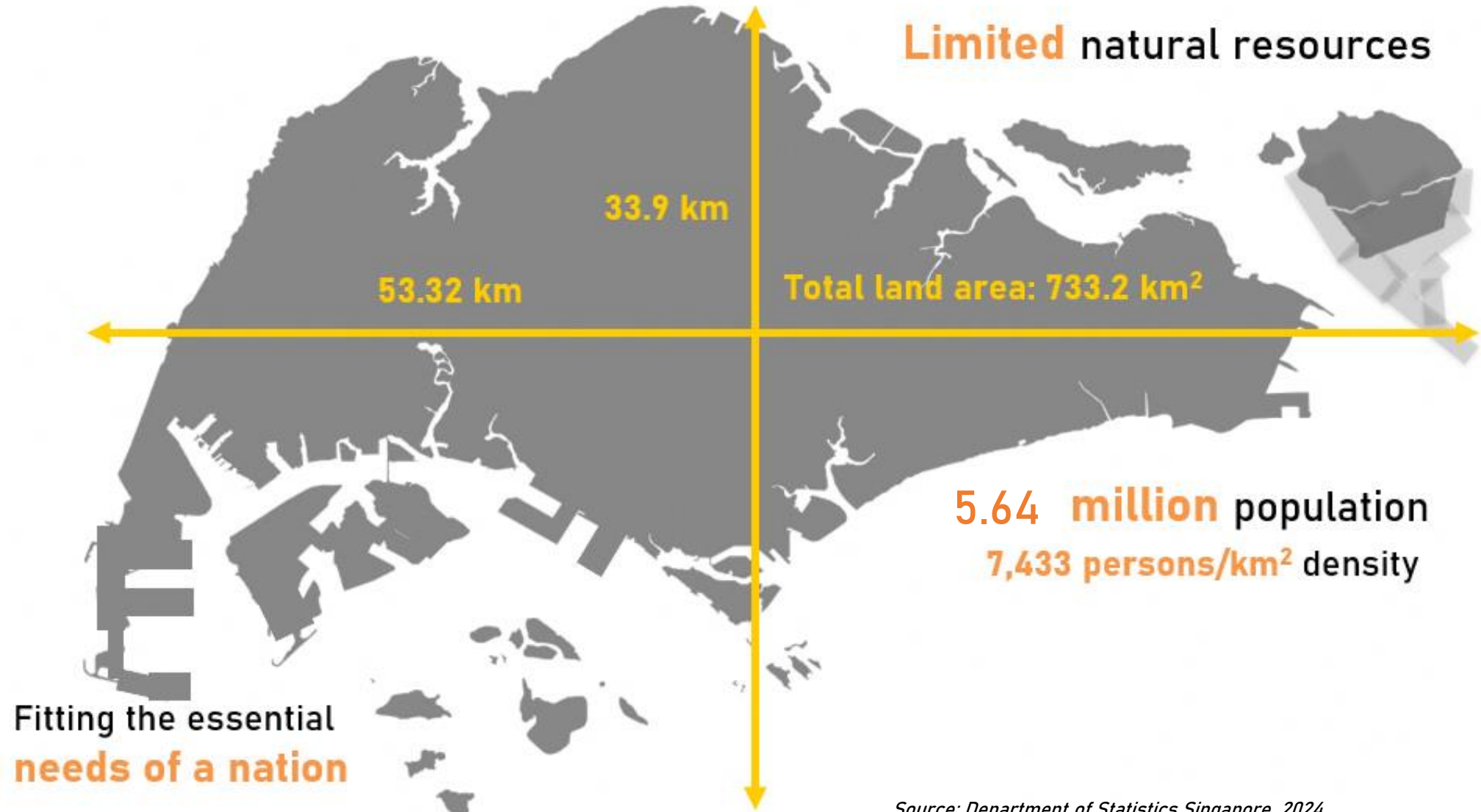


Ong Chong Xuan, **Joshua**
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Teaching Assistant, NUS Cities
BA (Hons) History, SOAS

The Liveability Framework (LF)

SINGAPORE A compact city-state's challenges



Source: Department of Statistics Singapore, 2024

Singapore in the 1950s & 1960s

Traffic congestion was one of the main challenges

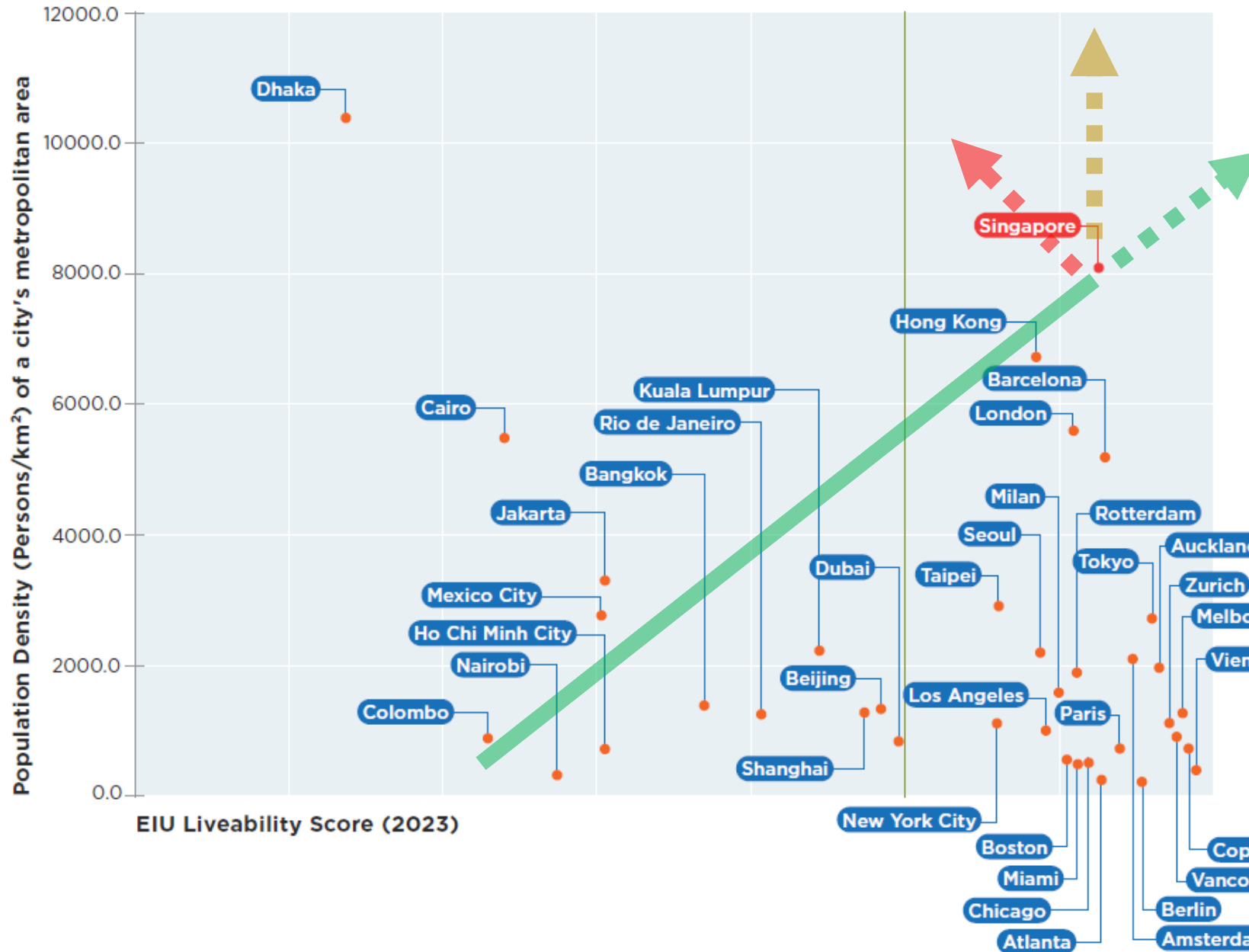


*Image source:
Assorted googled
media, 2025*

Singapore today



Image source: Assorted googled media. 2025



Liveability matrix

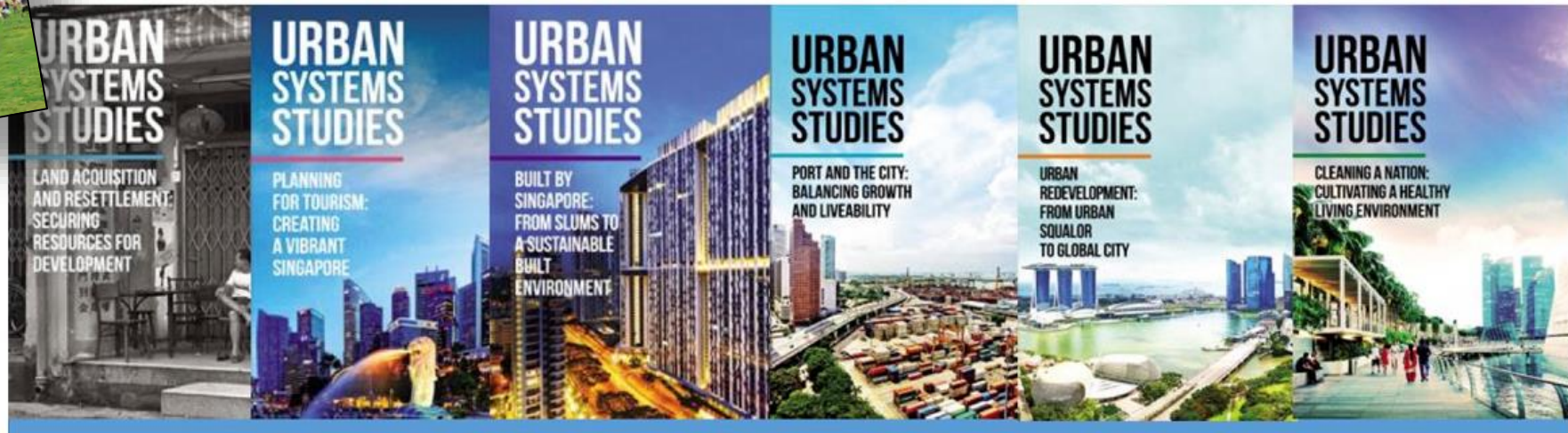
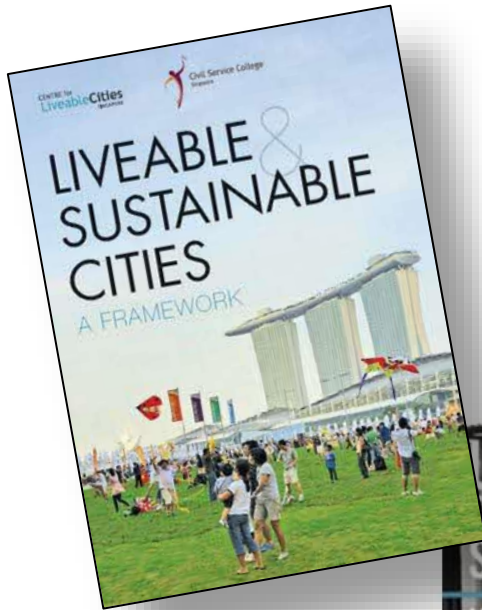
Singapore is a highly liveable and sustainable city

The CLC Liveability Matrix plots cities' population densities against liveability scores derived from The Economist Unit's (EIU's) Global Liveability Index; Singapore is one of a few high-density cities that scored well in EIU's Global Liveability Index

Image from Centre for Liveable Cities

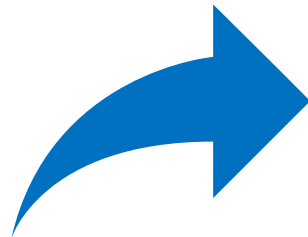
How did Singapore achieve liveability & sustainability?

- The Centre for Liveable Cities (CLC) distilled the principles from intensive research
- Cases have been comprehensively documented as **Urban Systems Studies (USS)**

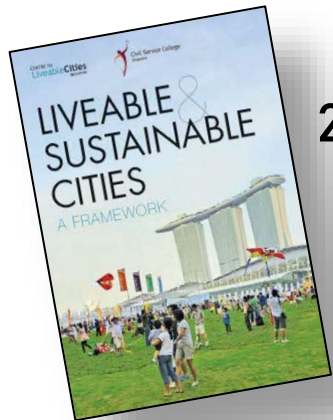


Liveability Framework for the Future

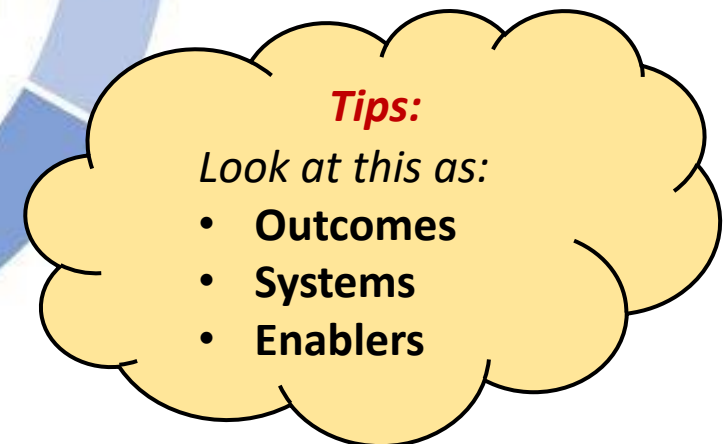
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2014

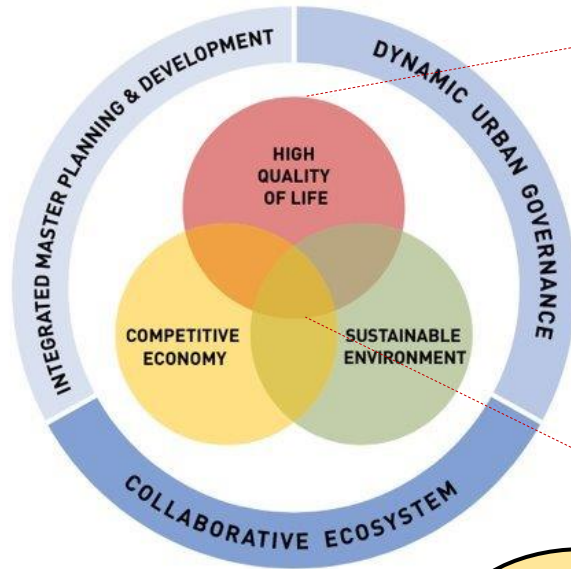


Liveability Framework for the Future



Outcome #1: High Quality of Life

The *What's*



Good mix and balance of:

1. Accessible, adequate and affordable public goods and services (*housing, transport, water, sanitation, health, etc.*)
2. Inclusivity, social cohesion, well-being, mental health and sense of belonging
3. Resilience and digital connectivity

Tips:

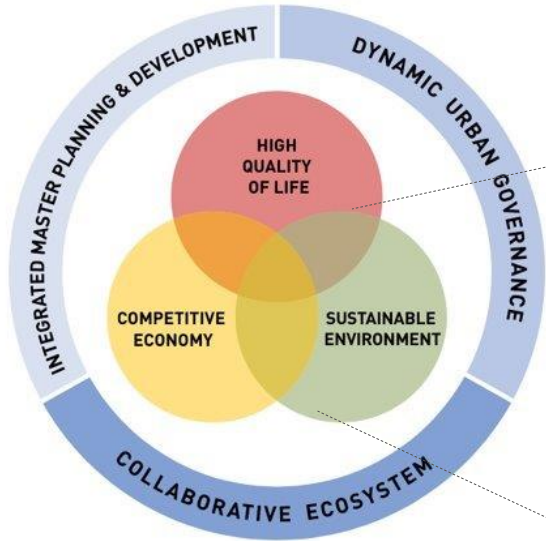
- Read and appreciate your case readings according to **what outcomes** are being achieved in Urban Systems Studies.
- **Connect** your project site discussions with the desired outcomes.

Seminar 1: Transport case

Seminar 2: Cities for All

Outcome #2: Sustainable Environment

The *What's*



Tips:

- Appreciate the themes/ideas shared by the experts in the Seminars, especially understand the **why's**.
- Discuss in your groups **what else** and **how else** the outcomes could be better achieved in your project sites.

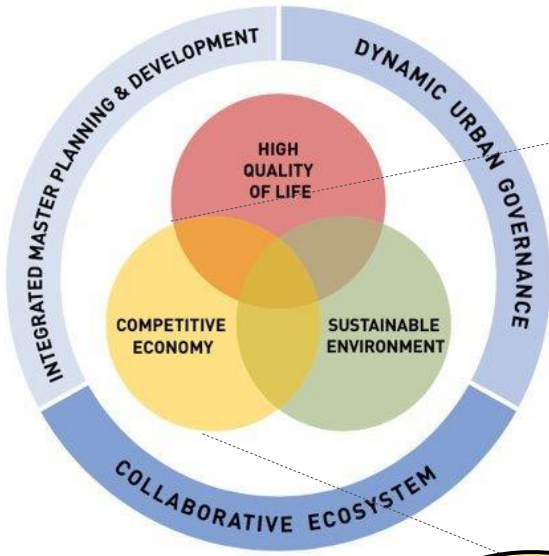
Good mix and balance of:

1. Clean, Good Quality Environment for Immediate and Long-term Resource Needs (*Air, water, drainage, waste, sewerage, noise*)
2. Holistic Green, Blue and Nature-based Eco-Systems (*Habitat, flora and fauna, nature reserves, etc.*)
3. Resilient to Risks (*climate change, natural disasters, food security*)

Seminar 3: Cities in Nature

Outcome #3: Competitive Economy

The *What's*



Good mix and balance of:

1. Competitive workforce: jobs, skills and employability
2. Attract talents, conducive business environment and inclusive work culture
3. Resilient economic growth

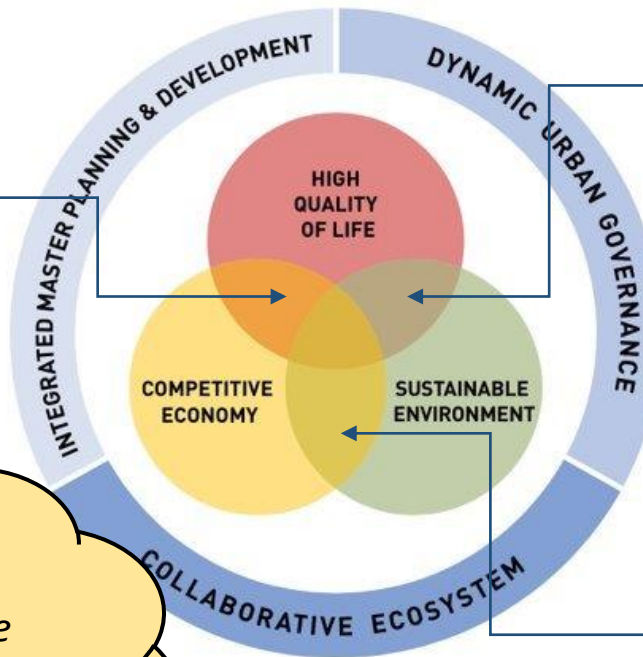
Tips:

- Appreciate the **complexity** of Urban Systems, their **challenges** and **interactions** to achieve the outcomes.
- Analyse the **stake-holders** and relevant **trade-offs** in your project sites.

Seminar 4: Smart Cities

Intersection of outcomes: Tensions & opportunities

- Redevelopment vs Conservation/Heritage
- Workforce productivity vs Health/Leisure
- Connectivity to support businesses vs Spatial quality



- Greenery vs Development
- Human – Wildlife conflicts

Tips:

- Project Site **problem prompts** are likely in these intersections.
- Use **systems maps**, **causal loops** and **stakeholder impact analysis** to uncover the issues

- Costs to businesses
- Cost of living

Three systems to achieve the outcomes

The *How's*

System 1

Integrated Master Planning & Development

Tips:

- Relate the **12 guiding principles** to Project Site.
- Look out for **applicable ideas** and **case examples** to cite.

Seminar 5: Planning & Governance

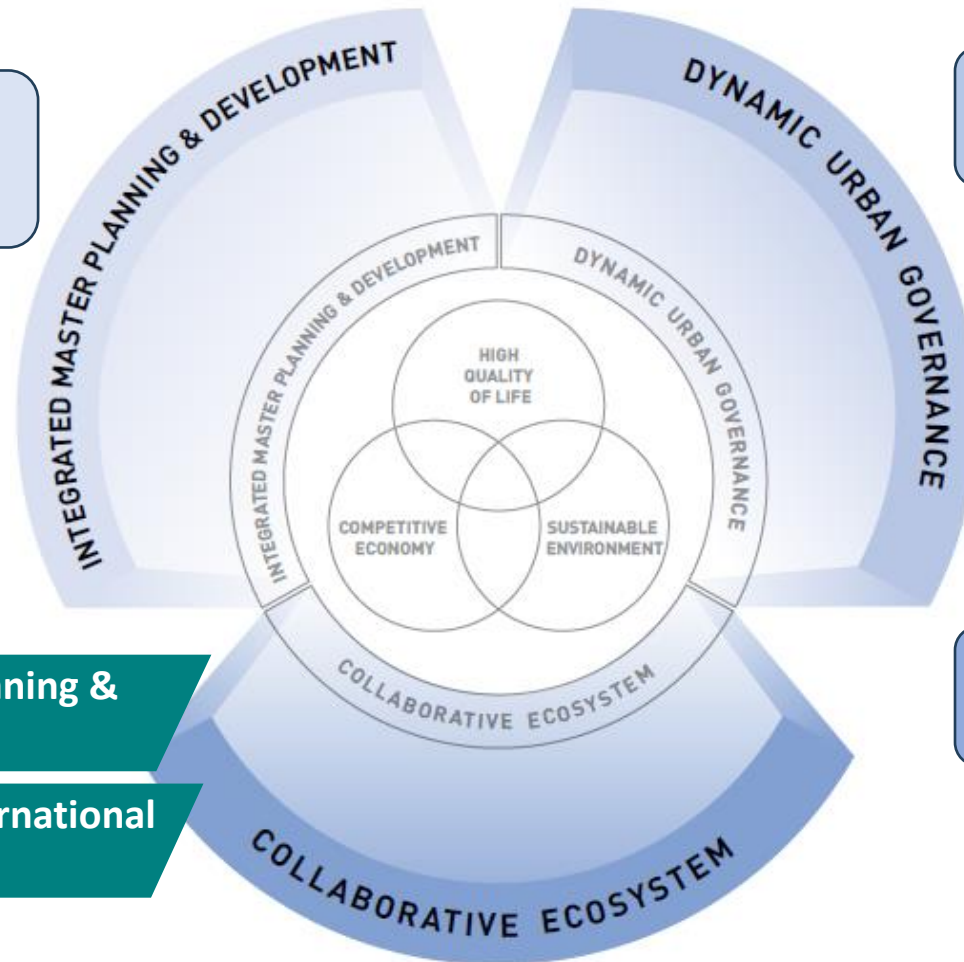
Seminar 6: International Case Studies

System 2

Dynamic Urban Governance

System 3

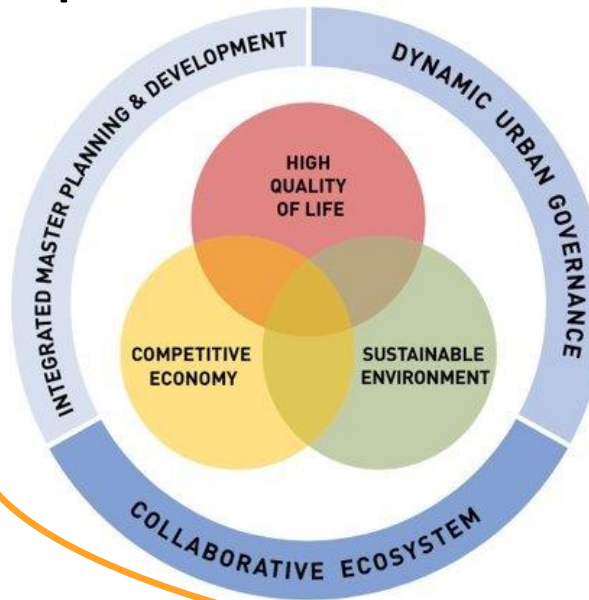
Collaborative Ecosystem



12 guiding principles in the 3 systems

Integrated Master Planning & Development

1. Think long term
2. Build for resilience
3. Seek win-win
4. Execute effectively
5. Innovate systemically



Dynamic Urban Governance

1. Lead with vision and pragmatism
2. Build a culture of integrity
3. Cultivate sound institutions
4. Engage and partner with the community
5. Work with markets

Collaborative Eco-system

1. Collective stewardship as a foundation for transformative collaborations
2. Forging strong international partnerships

Tips:

- In your Project Site, **relate your solutions** to some of these 12 guiding principles.
- Mention how these are being applied in your solutioning.

Enablers: Data and complexity science



In the LF diagram, we can think of “**enablers**” as **lubricants** operating within the space between the systems and outcomes. They **enhance the efficiency and effectiveness** of liveability outcomes delivery.

Image source: Centre for Liveable Cities, 2025

- Growing research to understand cities as **complex systems**
- Adopting **science of cities** approaches help us to understand **patterns, behaviours** and **emergence**
- Using **data-informed** planning and analytics to help provide **analysis, insights** and **foresights** in decision-making
 - Urban heat mitigation, sea level and flooding, ecological profiling, urban kampong research, pandemic responses, spatial impact analysis, activity-based mobility, X-minute cities, etc.

Tips:

Cite relevant **data-informed findings** in your analysis for your Project Site.

This approach has resulted in many innovations

For example...

Closing the water loop
with **NEWater, Urban Reservoirs, ABC Waters** etc.

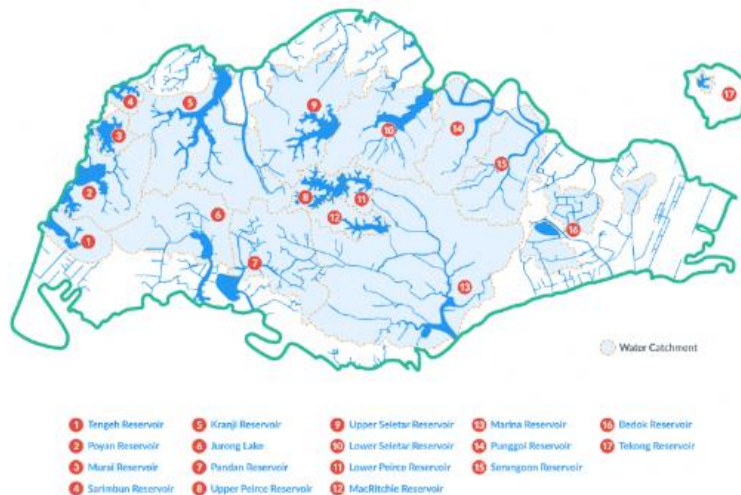
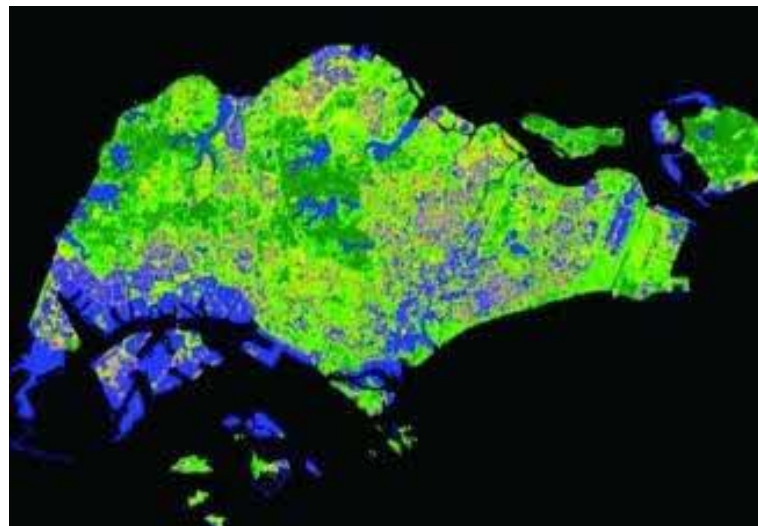


Image source: Centre for Liveable Cities, 2025

Almost 50% green cover
from **greening in tandem**
with **urban development**



Housing 80% of population
in **highly dense but liveable,**
socially integrated and
sustainable towns





Mid-point check...



CDE 2501:

- 6 Seminars, 7 Studios
- One Project Site, two Group presentations
- Individual quiz, attendance and performance

Liveability Framework (LF)

- 3 Outcomes
- 3 Systems (12 Guiding Principles)
- Enablers: Data and complexity science

Transport Case Study

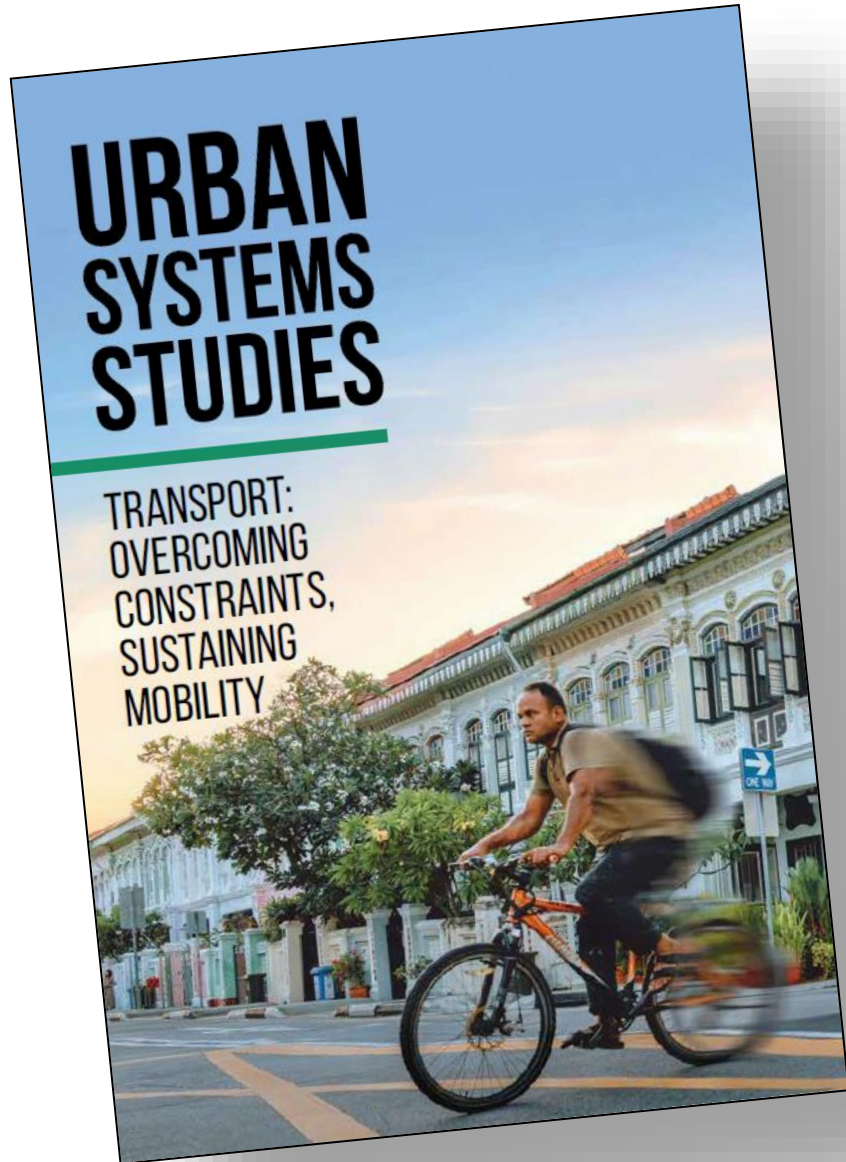


Image source: Centre for Liveable Cities, 2025

Urban systems approach to address Singapore's transport challenges

- Why is transport so complex?
- How do we solve it?
- What are some case examples?
- What's next?

Why is transport so complex?



A simple movement

- Point A to point B
- Single purpose
- Uni-direction
- Single pathway

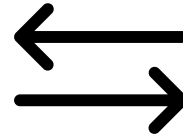


Image source: Straits Times, Singapore, 2025

An example of movements in a city

Commuting between home and workplace

Points A

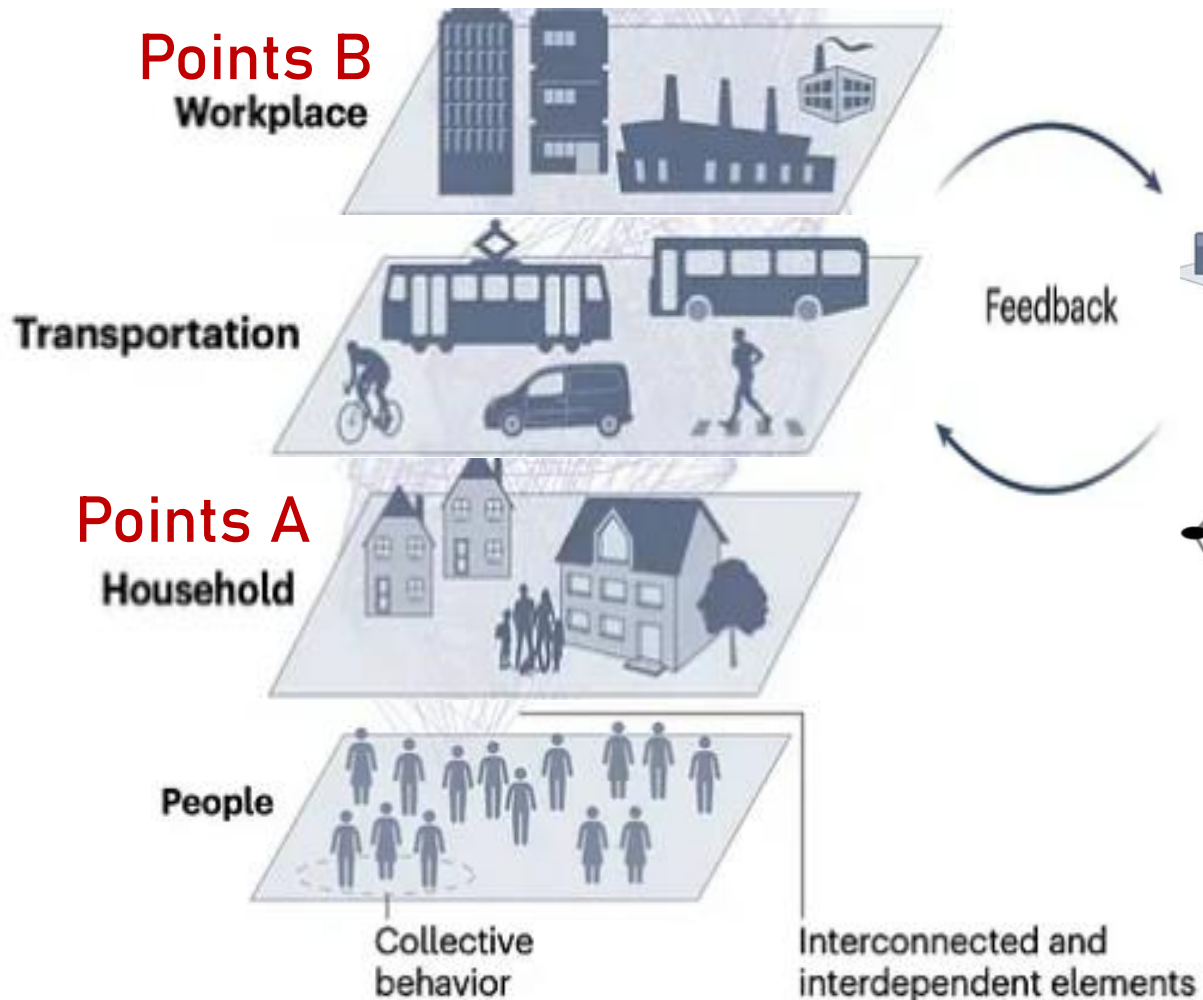


Workplace

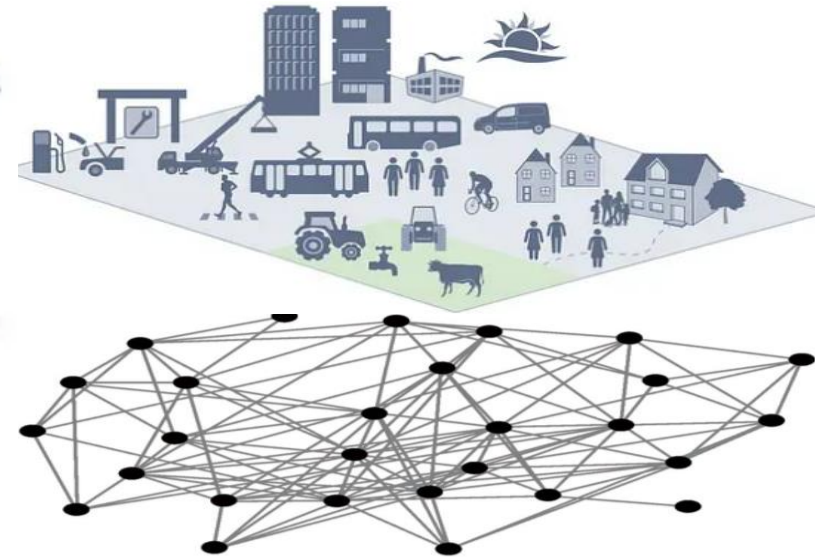
Points B



Complexity in movements



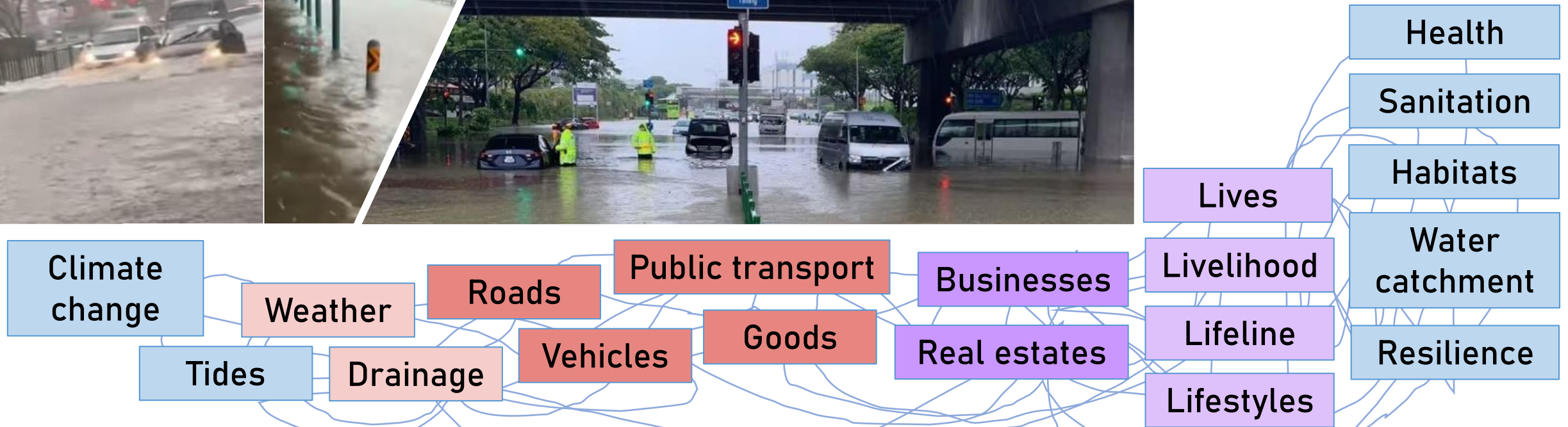
Network Effects



- Congestion
- Disruptions to network
- Peak and off-peak variations

Complexity: How can we better sense, analyse, understand, model and simulate, manage, price such complex movements to achieve the liveability outcomes?

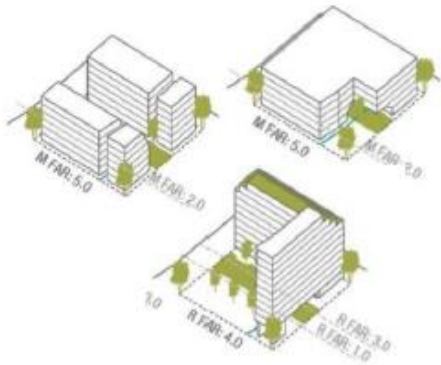
Zooming out, another example: Flooding in cities



In short, the city is a complex system

Solving urban problems is a complex endeavour that requires systems approach

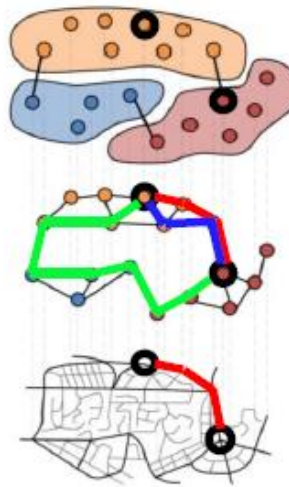
Land Use & Development Control



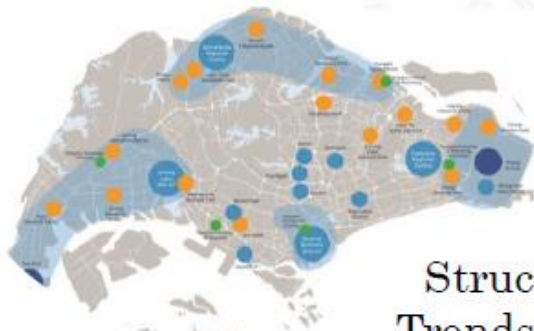
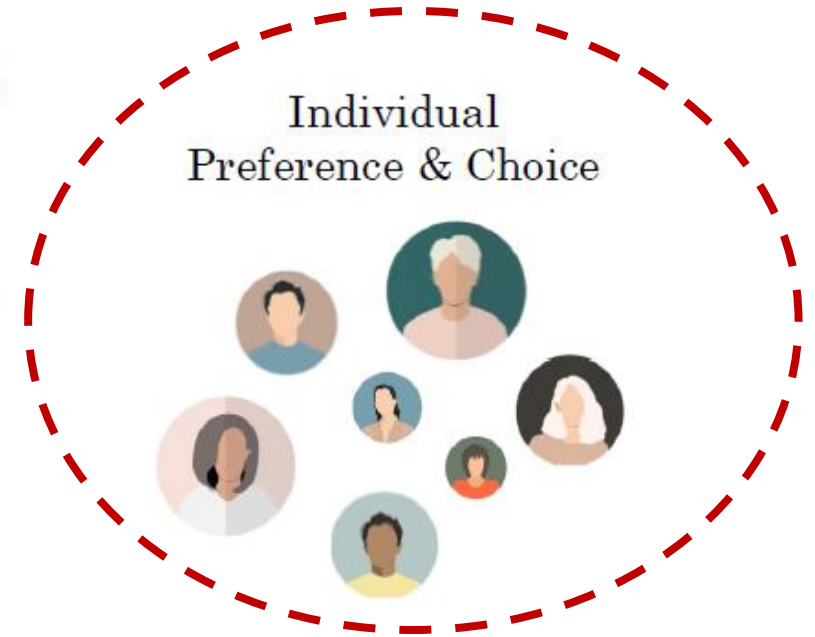
Activity & Programmes



Travel Demand & Mobility Options



Individual Preference & Choice

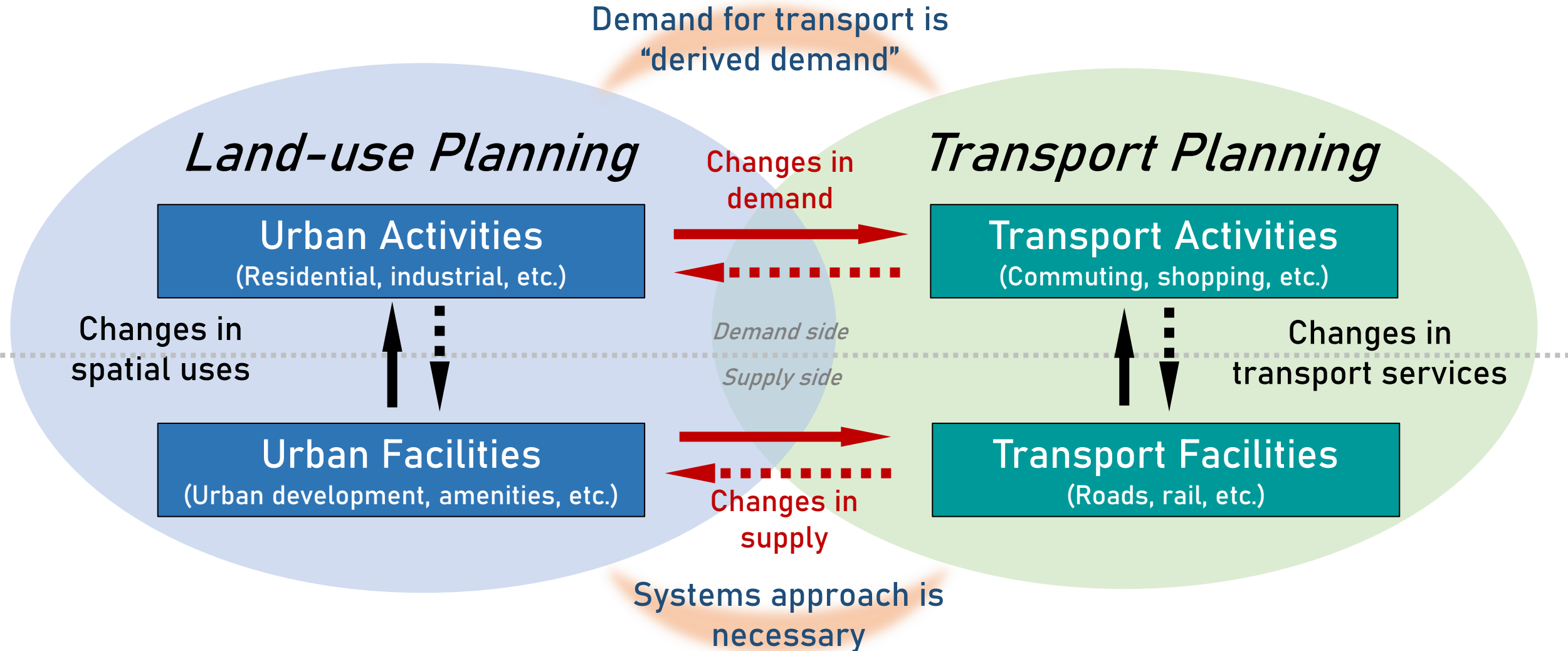


Structured Plan, Trends & Scenarios



Flows of People & Goods

Unpacking complex interactions



Source: Adapted by Looi TS from International Association of Traffic and Safety Science (IATSS) commemorative publication, 2015

How do we solve it ?

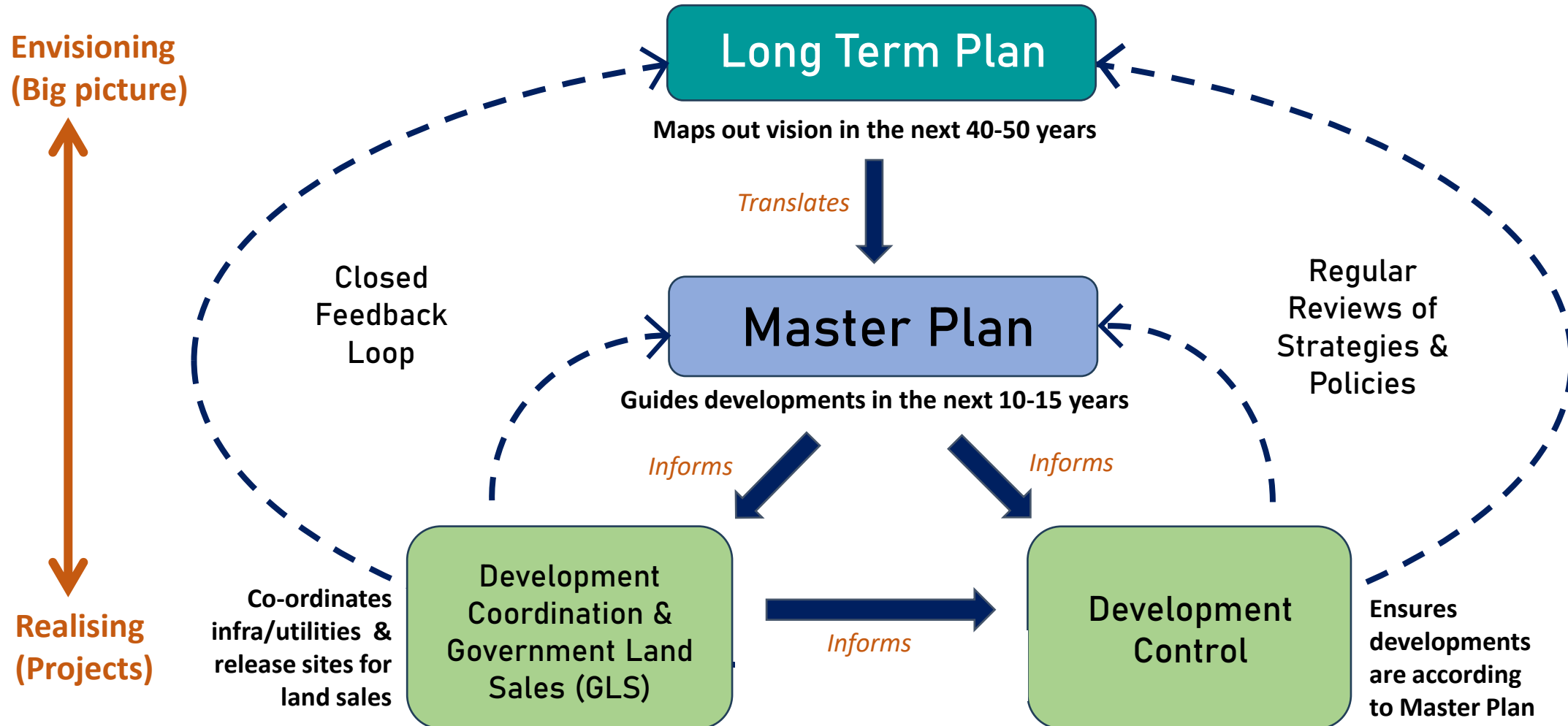
- (1) Planning and development
- (2) Public transport
- (3) Private transport
- (4) Value of data

How do we solve it ?

(1) Planning & development

Understanding integrated planning process

Linking “think long-term” with “think implementation”



Source: Adapted by Looi TS based on URA info, 2025

Early developments came with challenges

Transport infrastructure was identified as a strategic development priority

- Rising travel demand
- Worsening traffic congestion
- Unreliable bus services



Overcrowding was not unusual and passengers hanging onto a bus was a common sight.



Photo credits: LTA Singapore, 2025

Land transport strategies in early years

Land scarcity meant that a systems approach was needed for problem-solving

- Integrate land use and transport planning
- Develop a modest road network
- Improve public transport
- Manage traffic to optimize road utilization
- Manage travel demand

Source: A P G Menon, Singapore, 2015



Ministry of National Development



Public Works Department

Registry of
Vehicles

Ministry of
Communications

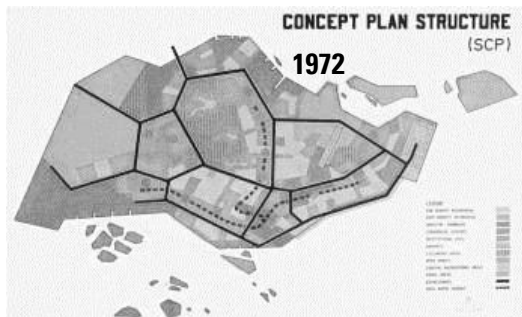
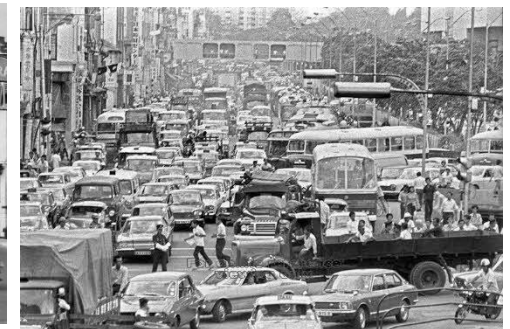


Image source: Assorted googled media, 2025



Today, we continue to adopt a systems approach

A mix of targeted pull-and-push intervention levers



Integrate planning & development

Integrate land-use and transport planning & development

Increase public transport usage

Promote and make public transport accessible, reliable & affordable



Encourage active mobility (cycling, walking)

Make active mobility inclusive and promote a healthy and liveable community



Reduce reliance on private transport

Facilitate car/ride sharing and mobility on demand



Adopt green transport

Promote sustainability across value chain and build a green eco-system



Source: LTA Singapore, 2025

First: Think long term to align with vision

Aspirations, competing demands, tradeoffs and resource constraints



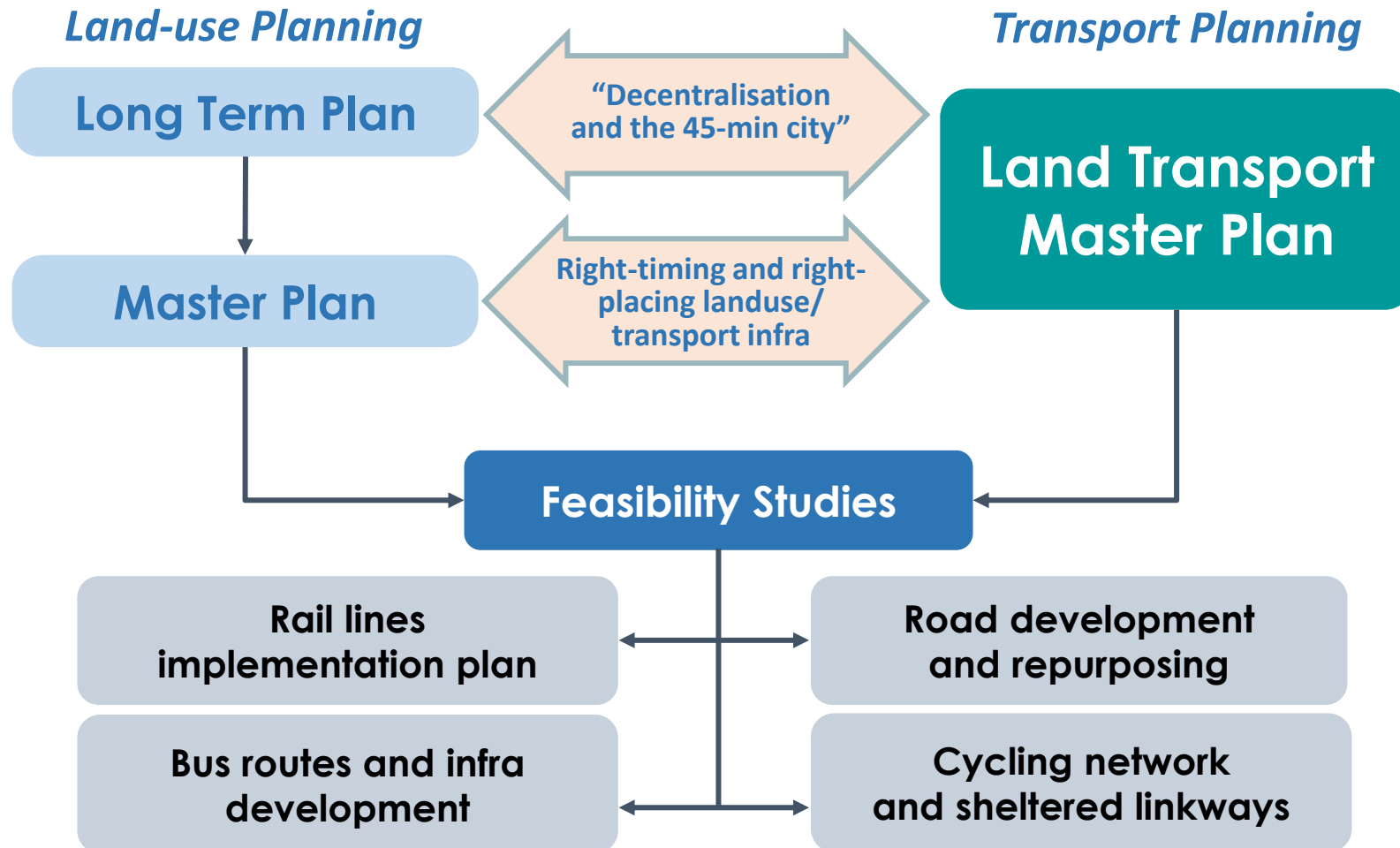
Source: URA Singapore 2021



Source: URA Singapore 2021

Second: Implement plans effectively

Clear plans and project priorities, premised on sound economics



Source: LTA Singapore, 2025

From broader plans....

Long term land use and transport plans (40-50 year horizon)

- Broad conceptual plans

Medium term land use and transport plans (10-15 year horizon)

- Staging plans
- Safeguard land needed

Rail, bus, WCR feasibility studies (5-10 year horizon)

- Technical studies
- Cost-benefit studies

To more specific ones

Third: Flexible and evolving master plans

From infra-led outputs to aspirational outcomes

20-Min Towns; 45-Min City

- 20 min town journeys
- 9 in 10 peak-period journeys less than 45 mins

Transport for All

- Gracious and caring commuting culture
- Barrier-free journeys

Healthy Lives; Safer Journeys

- More spaces for active mobility
- Cleaner energy

Source: LTA Singapore, 2025

Early developments were infra-led, guided by State and City Planning 1971, the Strategic Transport Plan and various studies



1996

A world class land transport



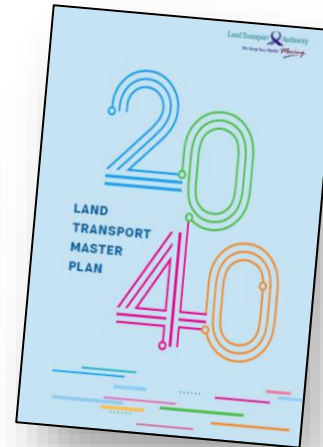
2008

Make public transport a choice mode



2013

A people centred land transport



2019

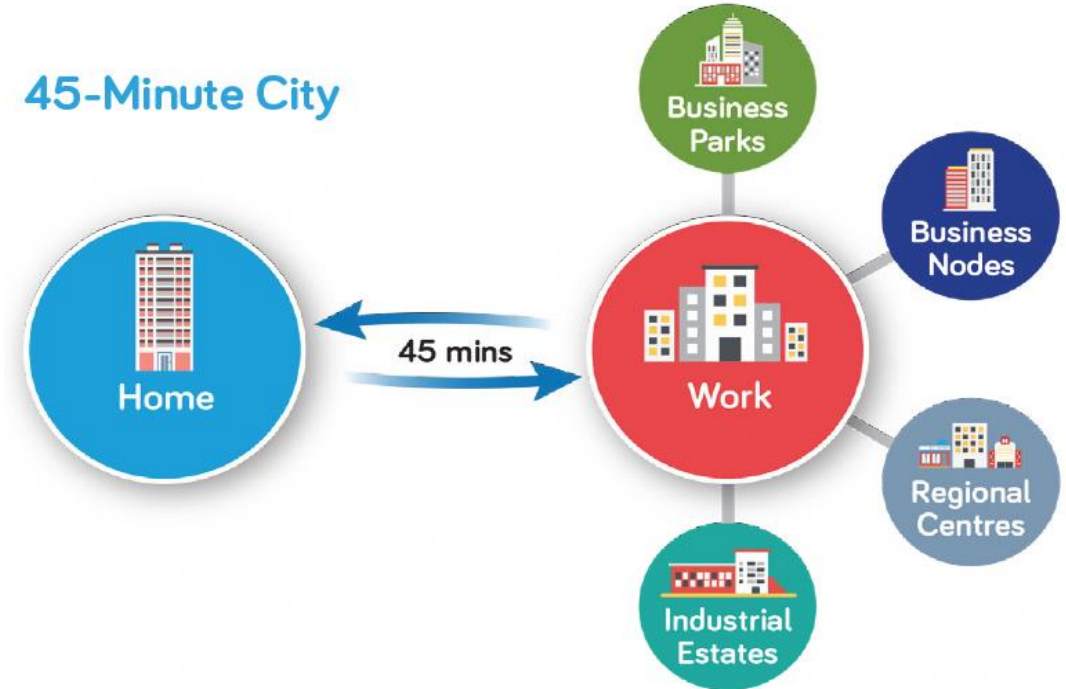
Bringing Singapore together

Envisioning a 20-min towns and a 45-min city

20-Minute Towns



45-Minute City

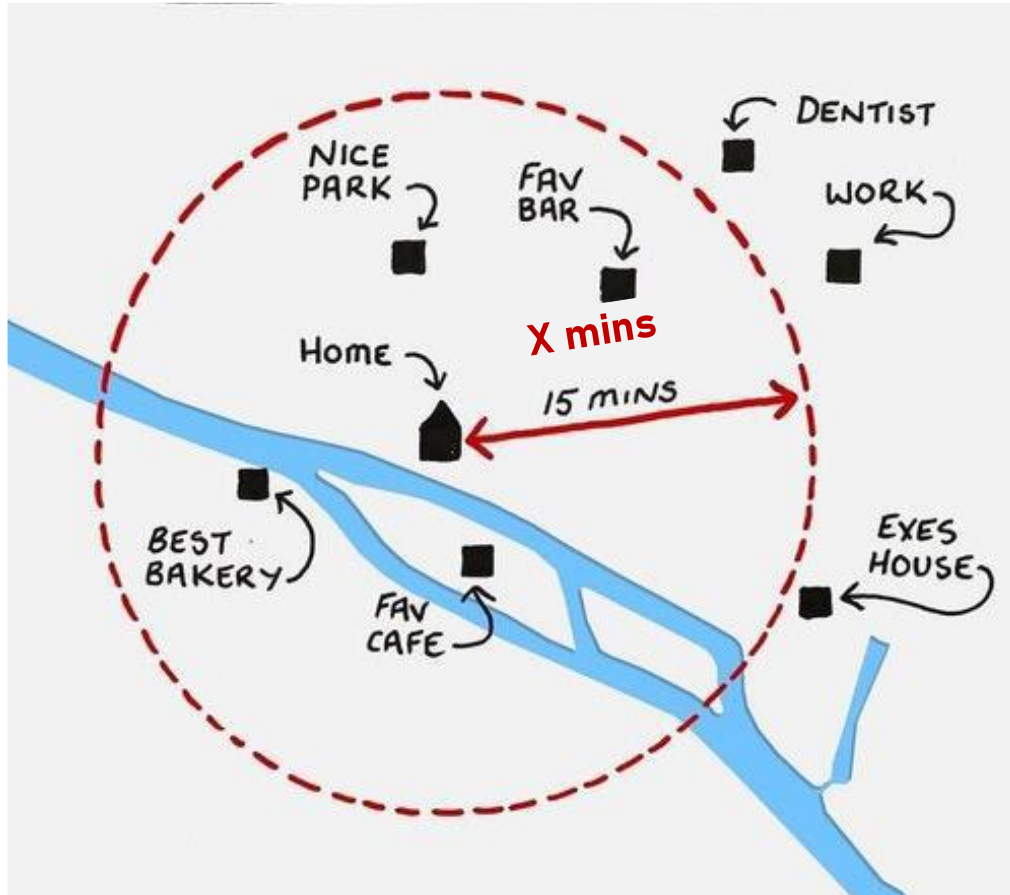


20-Minute Town: All journeys to the nearest neighbourhood centre on Walk-Cycle-Ride are completed in less than 20 minutes

45-Minute City: 9 in 10 peak-period journeys on Walk-Cycle-Ride are completed in less than 45 minutes

Adopting a “X-min city” model

Visionary target set using **travel time budget (TTB)** as a policy



Source: Phang SY Singapore 2021

Recent impetus:

- Digital revolution
- Pandemic lockdowns and work-from-home (WFH) practice
- Climate change agenda
- Adopted by Global Network of Cities for Climate, the C40 Cities in May 2020

Some examples: 2021 Scotland 20-min neighbourhood; 2020 Paris 15-min city; 2018 Melbourne 20-min neighbourhood; 2016 Sydney 30-min city; 2016 Shanghai 15-min community-life circle; 2008 Portland 20-min neighbourhood

Unpacking accessibility

Target

Accessibility

Destinations or goods and services that can be reached or being served

Means

Mobility

- Transport policy
- Transport capacity
- Integration
- Transit operations
- Ride hailing services
- Modes and splits

Proximity

- Neighborhood
- Jobs and housing spatial balance
- Monocentric or polycentric urban spatial form
- Work from home/anywhere

Connectivity

- Broadband penetration
- Smartphones
- Transit mobility apps
- Ride hailing platforms
- Trip saving delivery services

Inter-dependance

Sources: Jonathan Levine et al, 2019, SY Phang 2021, with adaptation by Looi TS, 2025

How do we solve it?

(2) Public Transport

Defining walk-cycle-ride (WCR) choices

Shared Transport

- Point-to-Point (P2P)
- On-demand
- Mainly “urgent” rides
- Commercially priced
- Competitive market

**WALK
CYCLE
RIDE**
SINGAPORE



Active Mobility

- First-mile-last-mile
- Micro-mode choice
- Some bike-sharing
- Regulated user behaviour
- Registration of e-bikes

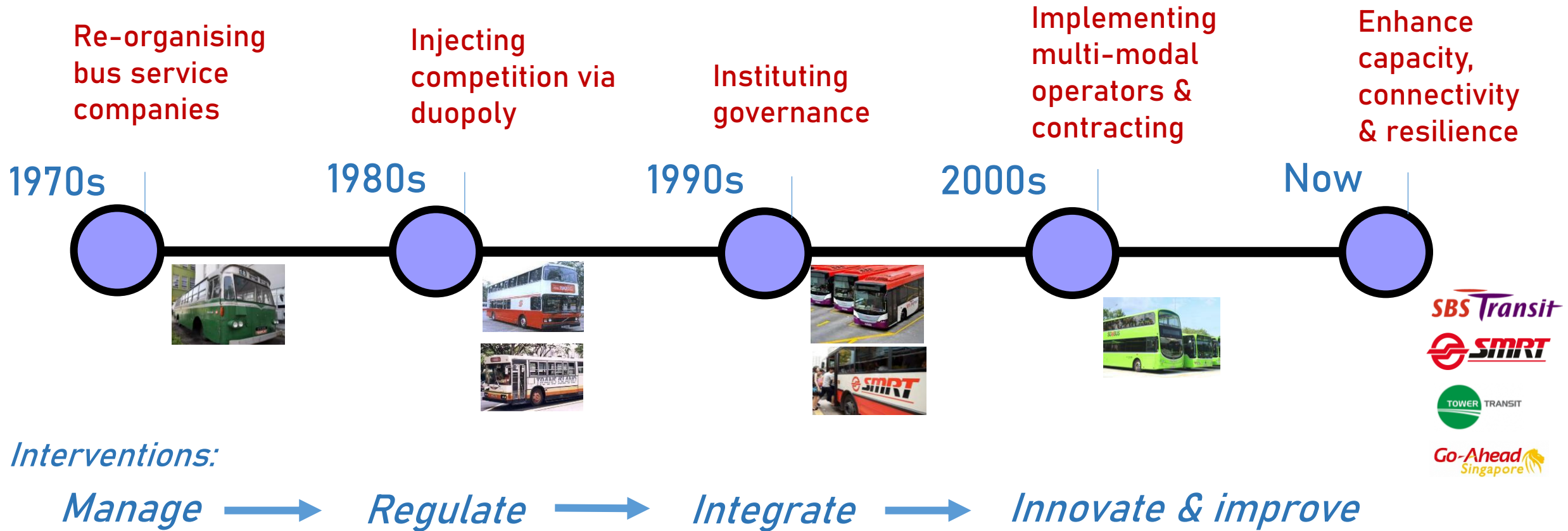
Mass Public Transport

- Mass commuting backbone
- Universal service obligations
- Affordable fares
- Promotion of desired behaviour
- Tightly regulated services

Source: LTA Singapore, 2021

First: Transform bus services

Consistent priority to improve bus services from the outset



Source: Looi TS, researched from various public sources, 2024

Second: Implement the MRT



Image credits: MediaCorp Singapore, Straits Times Singapore
Permission required for reproduction

Both sides of the picture

The MRT Debate
-6 OCT 1980 Ch 5, 10.10 p.m. tonight

Chaired by Mr Liu Thai Ker, Chief Executive Officer, HDB
Members: Kenneth Hansen and Prof John Kain (Harvard Review Team)
— speaking for the all-bus system
Bruno Wildermuth and Robert Hubbard (Wilbur Smith and Associates)
— speaking for the MRT

Milton Tan, President, Automobile Association of Singapore
Joseph Yee, Senior Executive Engineer, PWD
Mah Bow Tan, Asst. General Manager, SBS
Wong Nan Jong, a General Manager of OCBC
Prof Bill Lim, Head, School of Architecture, NUS
Toh Thion Ser, Associate News Editor, Straits Times

MRT system

By ABDULLAH TARMUGI

WORK on the MRT must start by 1984 at the latest if it is to be completed by 1992. Work should start now if the GDP grows at 8 per cent per annum.

An all-bus system cannot take more than 260,000 people working in the Central Area. And it has been projected that this number will exceed this by 1992.

An all-bus system will mean:

- Buses will crawl bumper to bumper. Streets will be choked with traffic.
- Bus-stops will overflow with commuters waiting to scramble through the only entrance of any bus that comes along.
- Passengers will be crammed in the buses. The ride will be hot and long.
- Travelling will be a breeze. The ride is smooth and in air-conditioned comfort.
- Fares will be comparable with the prices of bus tickets. Buses will not be cheaper to run, what with oil prices perpetually rising.
- With MRT, the Central Area can be allowed to grow, making Singapore into the global city it is meant to be.
- The MRT is expensive to build, but look at the benefits.

All-bus system

SINGAPORE does not need an MRT, especially an underground one. There are better and cheaper options: An all-bus system, a one-line MRT and the "people-movers".

Buses will run on a "hub and spoke" network and can cope with more than 300,000 workers employed in the Central Area.

With the all-bus system:

- Large, high performance buses will ply non-stop between the Central Area and between the new towns.
- Buses can run at 90 km/h along the expressways provided other vehicles are kept off these routes.
- There'll be fewer stops in the Central Area so that travel will be faster and the buses can go at faster speeds.
- Buses can run two abreast on two bus lanes instead of one.
- Capital and operating costs will be cheaper and the system can be started immediately.
- For the bus system to work, metering devices and tolls will have to be introduced at entry points to the expressways, an ALS would have to be imposed and the peak period will have to be stretched from 1 1/2 to 2 hours.
- The other options are the one-line rail-based MRT and the "people-movers".
- The rail system will only be an East-West line minus the North-South line. This makes it cheaper than the full MRT.
- This limited rail system can also be above ground, making it cheaper.
- An elevated rail system in the Central Area is not visually offensive.
- Although the system can take in workers in the Central Area, this area must continue to grow to the point, Singapore decentralise.

K. HANSEN
KENNETH HANSEN

R. HUBBARD
ROBERT HUBBARD

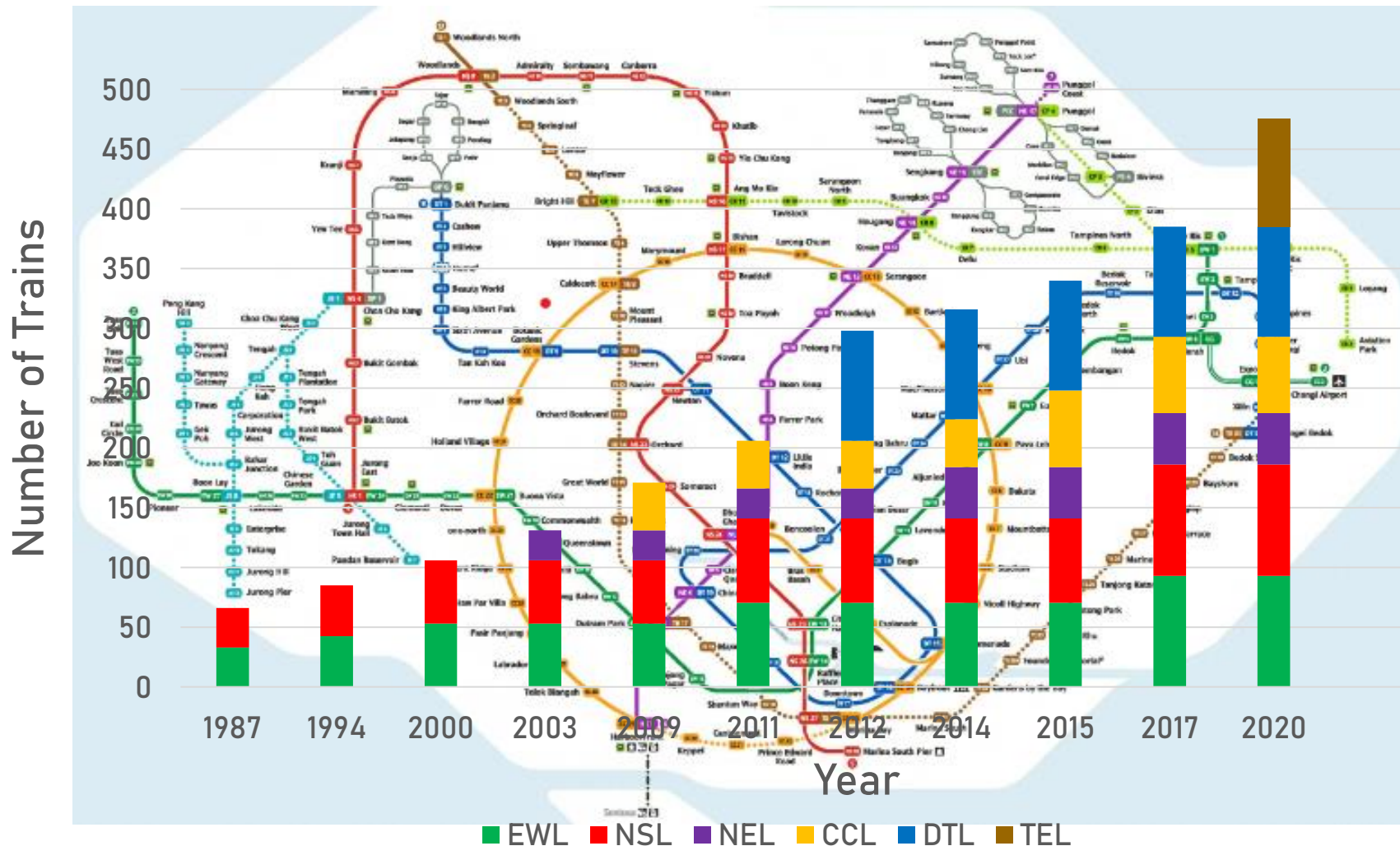
L. THAI KER
LIU THAI KER



The moderator of the debate, Mr Liu Thai Ker (centre), with a group of Q&A panelists and the two expert teams

Build a comprehensive network of rail lines

As the **backbone** of public transport



• **1987:**

1st MRT line
5 stations
6 km

• **2020:**

3 LRT lines
41 stations
28.6 km

6 MRT lines
122 stations
203 km
4 lines: **Fully Automated**

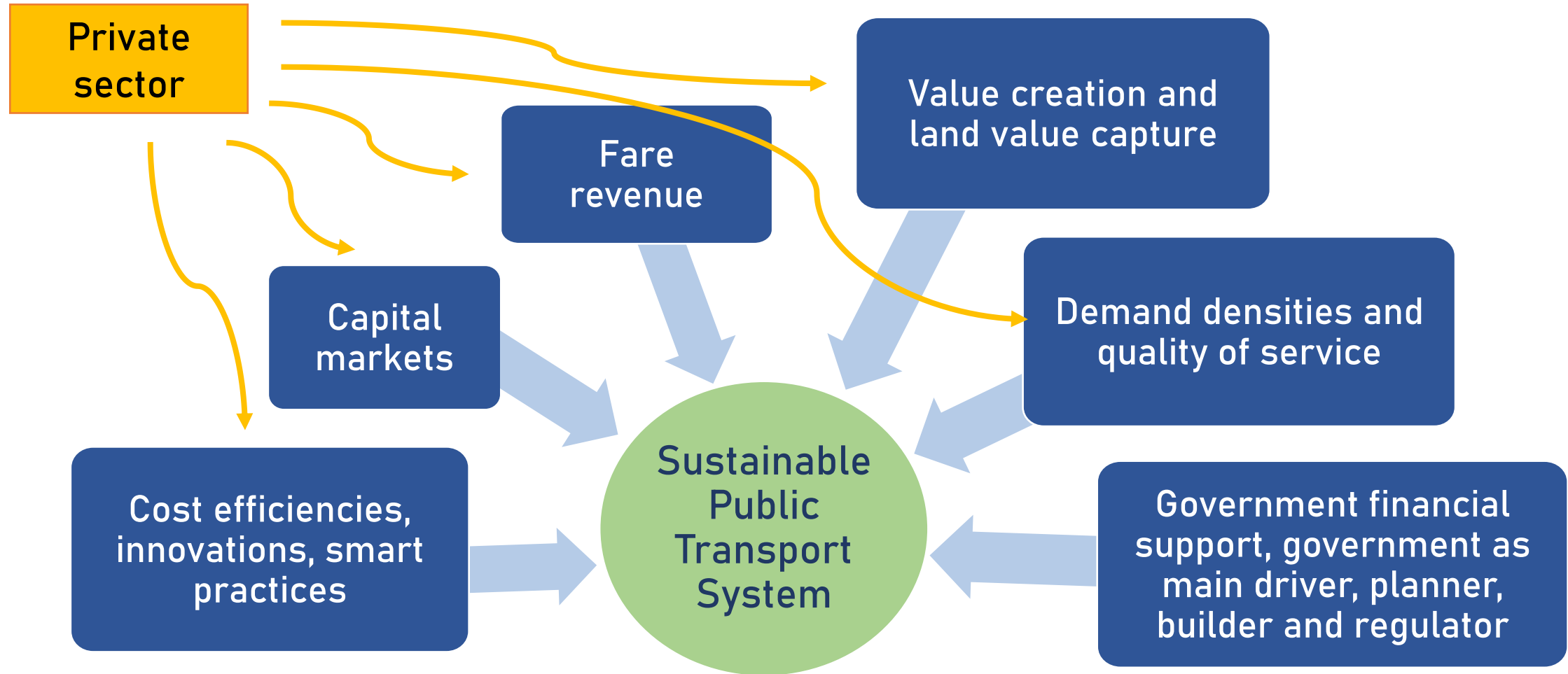
• **By 2030:**

8 MRT lines
360 km

Source: LTA Singapore 2023

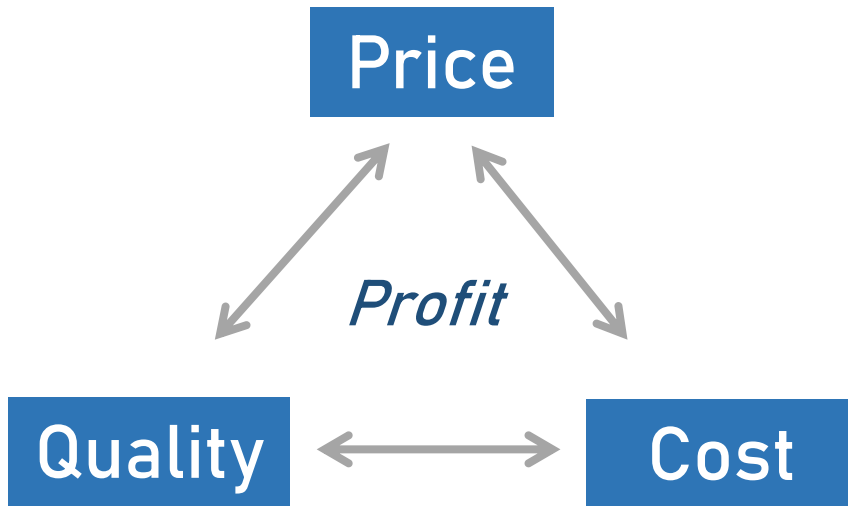
Third: Work with market (public-private partnership)

Bring in the merits of having private sector to deliver outcomes

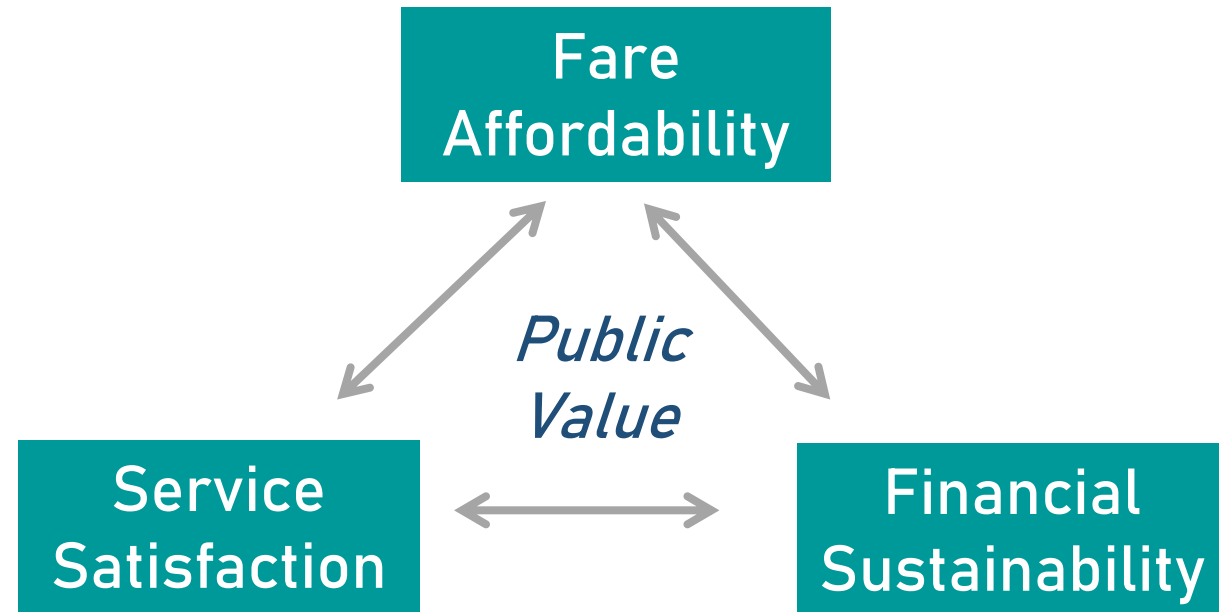


Fourth: Instituting regulatory roles

Firm in a market



Roles of government



Source: Looi TS, researched from various public sources, 2025

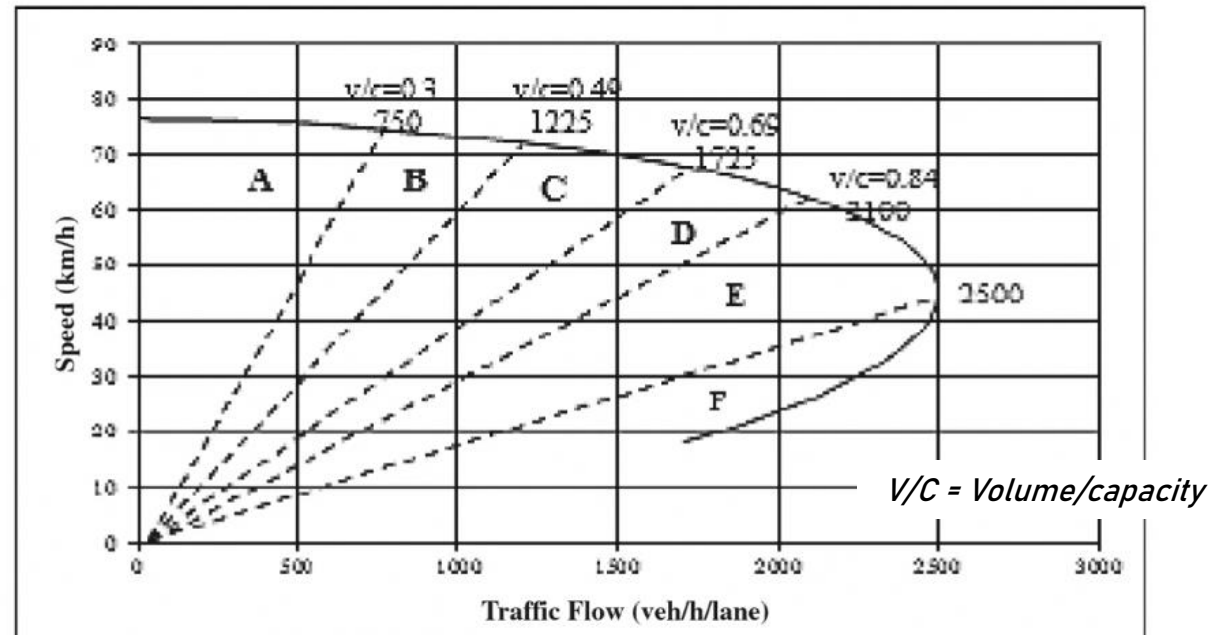
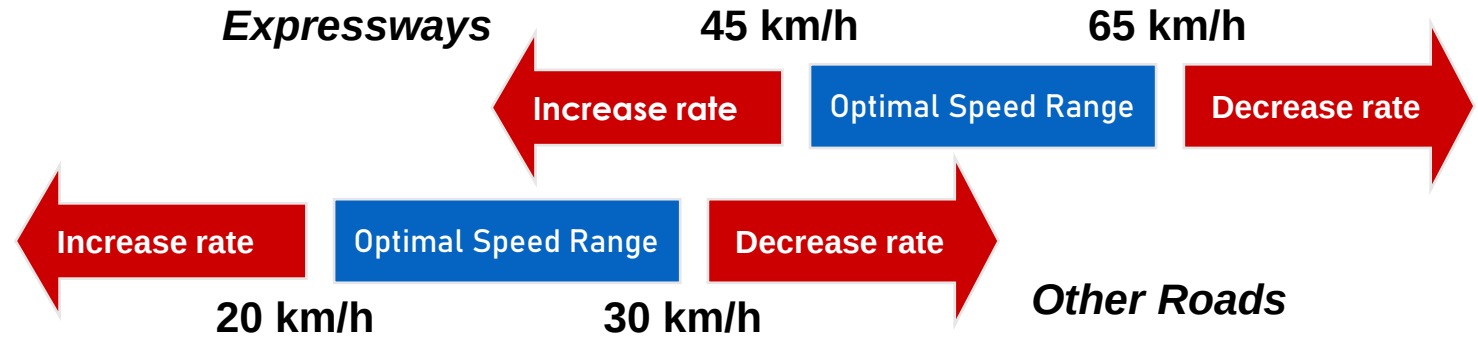
How do we solve it ?

(3) Private Transport

Pricing externalities: Road congestion



Source: LTA Singapore, 2025

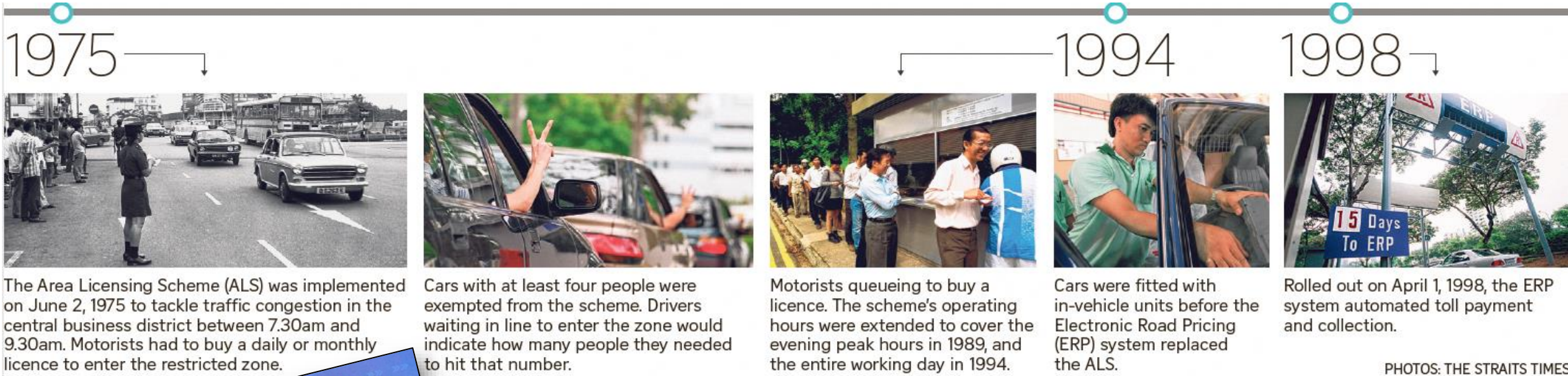


Source: Public Works Department Singapore

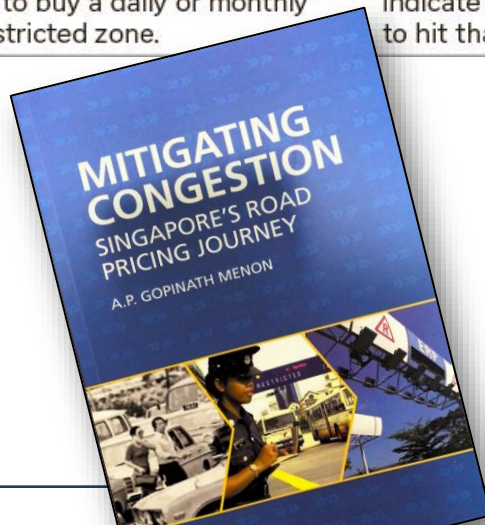
Fig. 28. Speed-flow graph for expressways.

Manage demands for road capacity

Potential to go beyond cordon and point pricing...



Source: ST 1998,
Permission required
for reproduction



Source: A P G Menon, Singapore, 2015

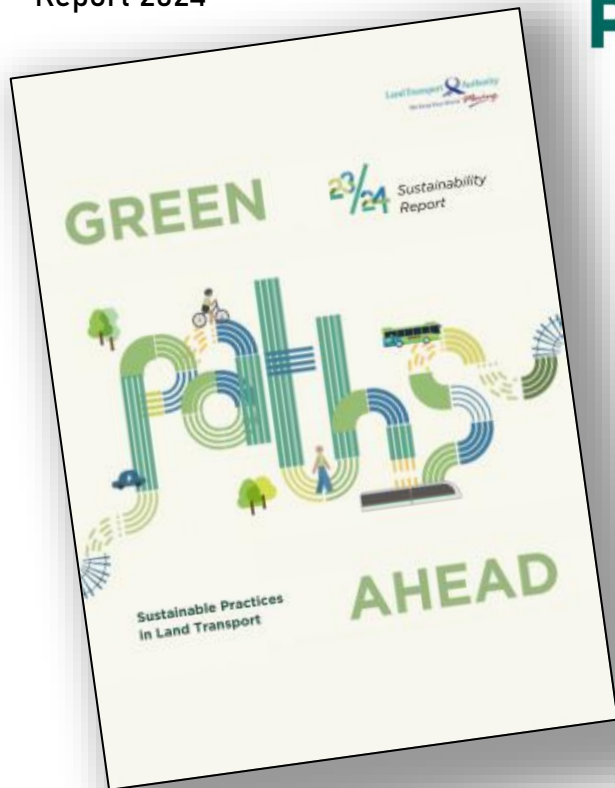
... to distance-based pricing

... coming up next-gen ERP

EV roadmap towards a net-zero vision

Aggressive targets set

LTA Sustainability
Report 2024



2050

Net-zero city

Key movers: Vehicle electrification & more WCR

2040

Cleaner energy vehicle vision

By 2040, **100%** vehicles on cleaner energy

2030

60,000 EV charging points

All new car registrations to be cleaner-energy



2025

No new diesel car registrations and
every HDB town an **EV-ready** town

Source and image credits: LTA Singapore, 2025

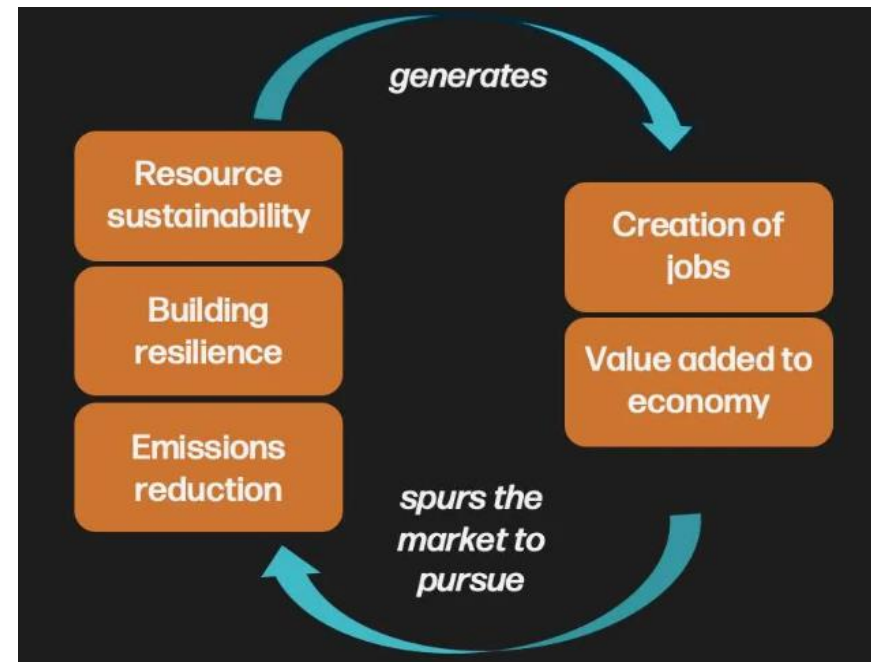
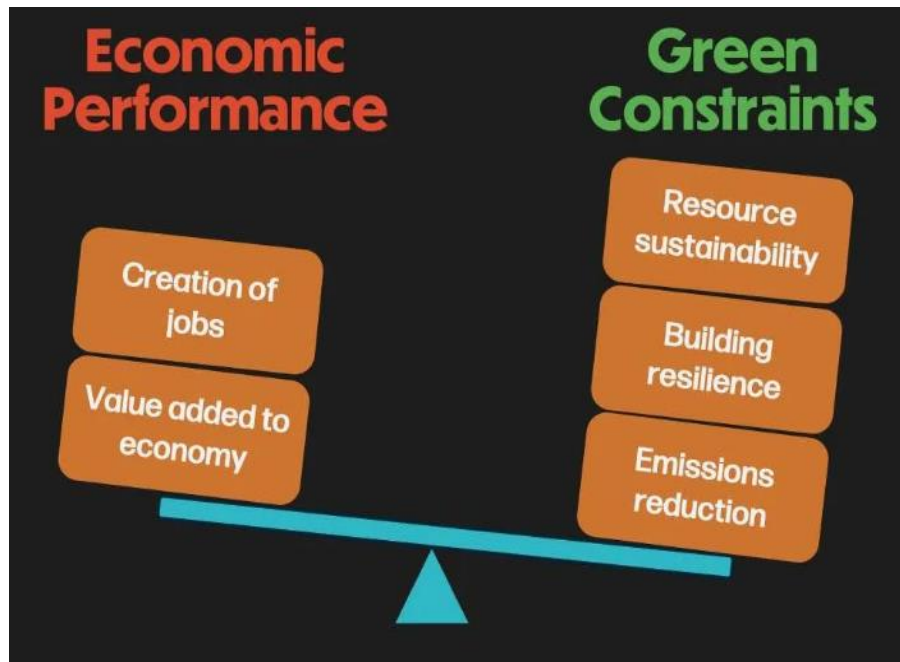
From trade-off to value capture

Opportunistic shift in response to climate change

From thinking of green constraints
as a trade-off of
economic development



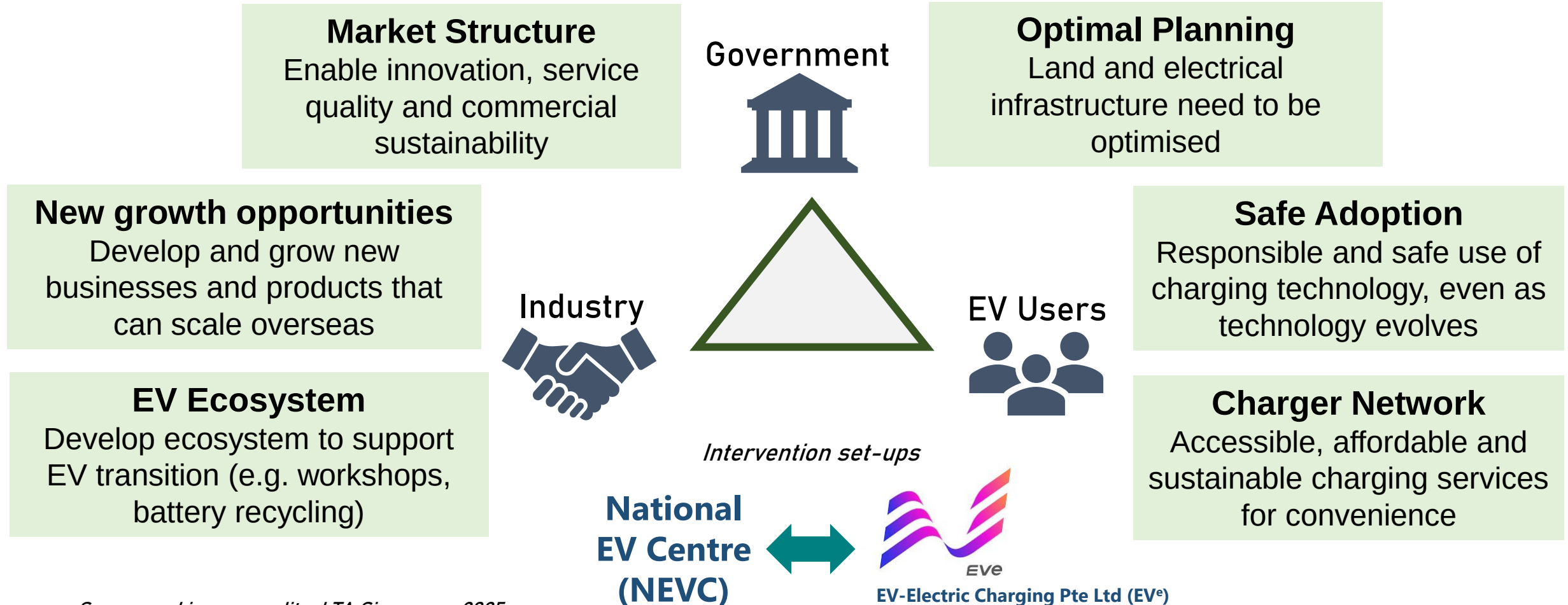
To harnessing
green opportunities for
economic growth



Source and image credits: e-module Learn PSD, Singapore, 2025

Today's EV planning and governance

Partnering public-private-people sectors

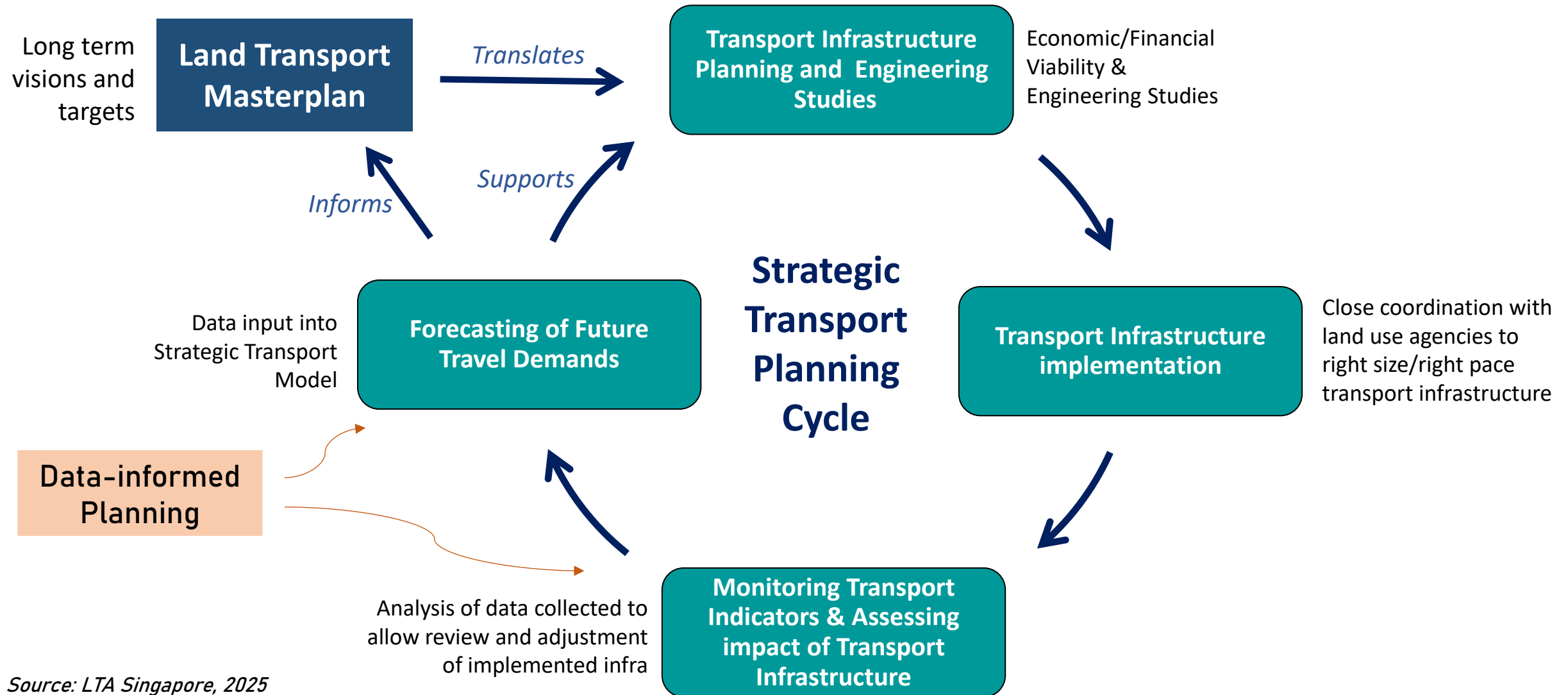


Source and image credits: LTA Singapore, 2025

How do we solve it ?

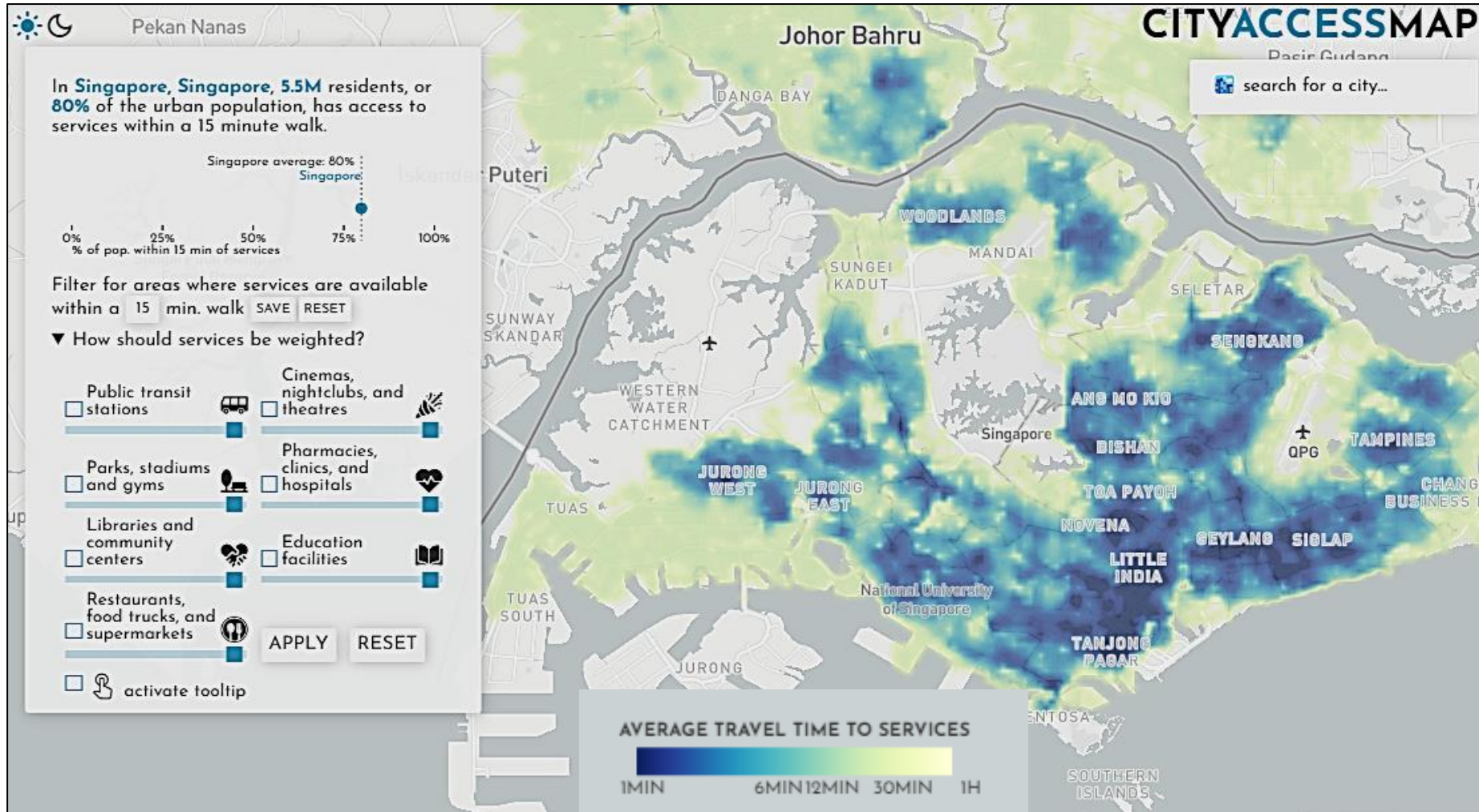
(4) Value of Data

Data-informed transport planning



Source: LTA Singapore, 2025

Data for complexity sensing & visualisation



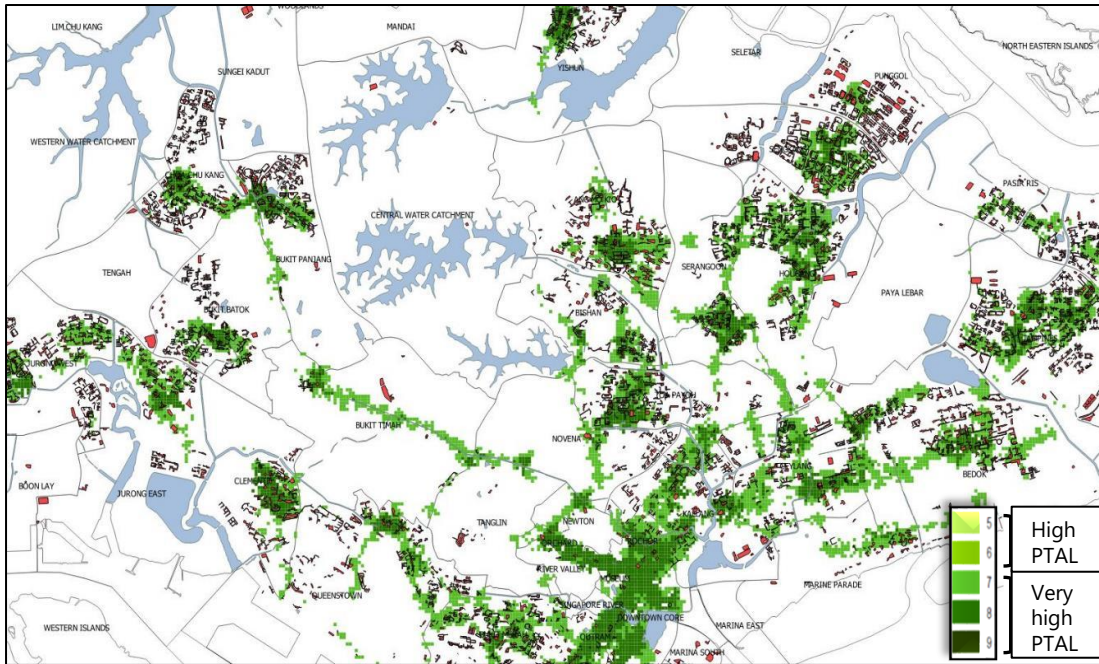
80% of residents (5.5mil) have access to services within a 15-min walk

- Transit stations
- Parks/stadiums/gym
- Libraries/community centres
- Restaurants/food centres/supermarkets
- Cinemas/theatres/clubs
- Clinics/hospitals/pharmacies
- Schools/education facilities

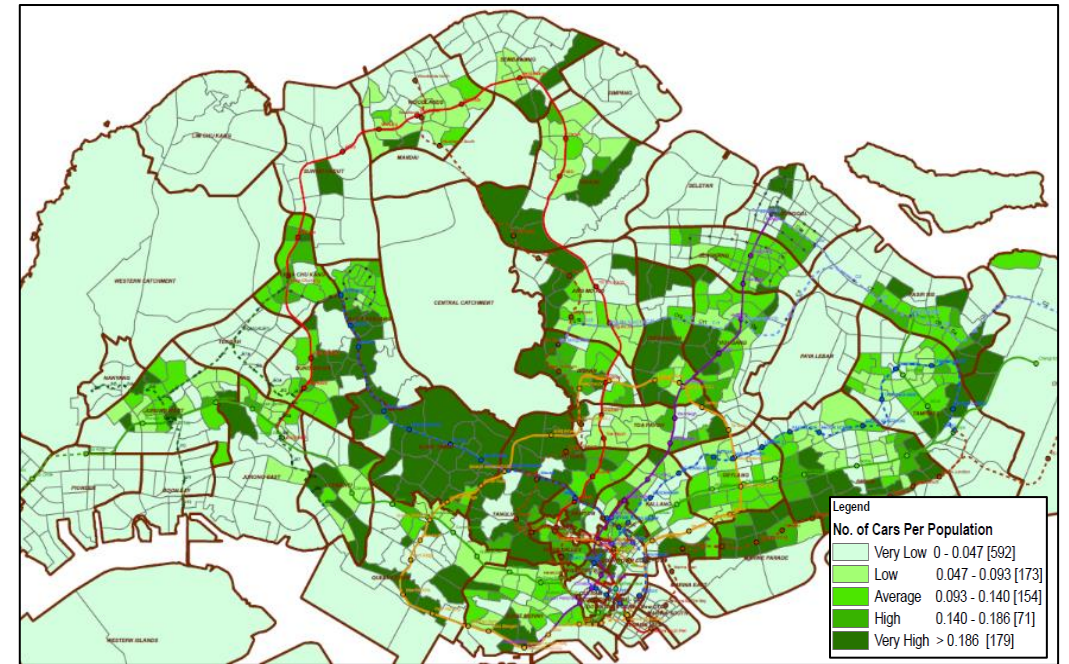
Source and Image credits: www.cityaccessmap.com, accessed by Looi TS, Jan 2025

Examining complex relationships

High PTAL areas



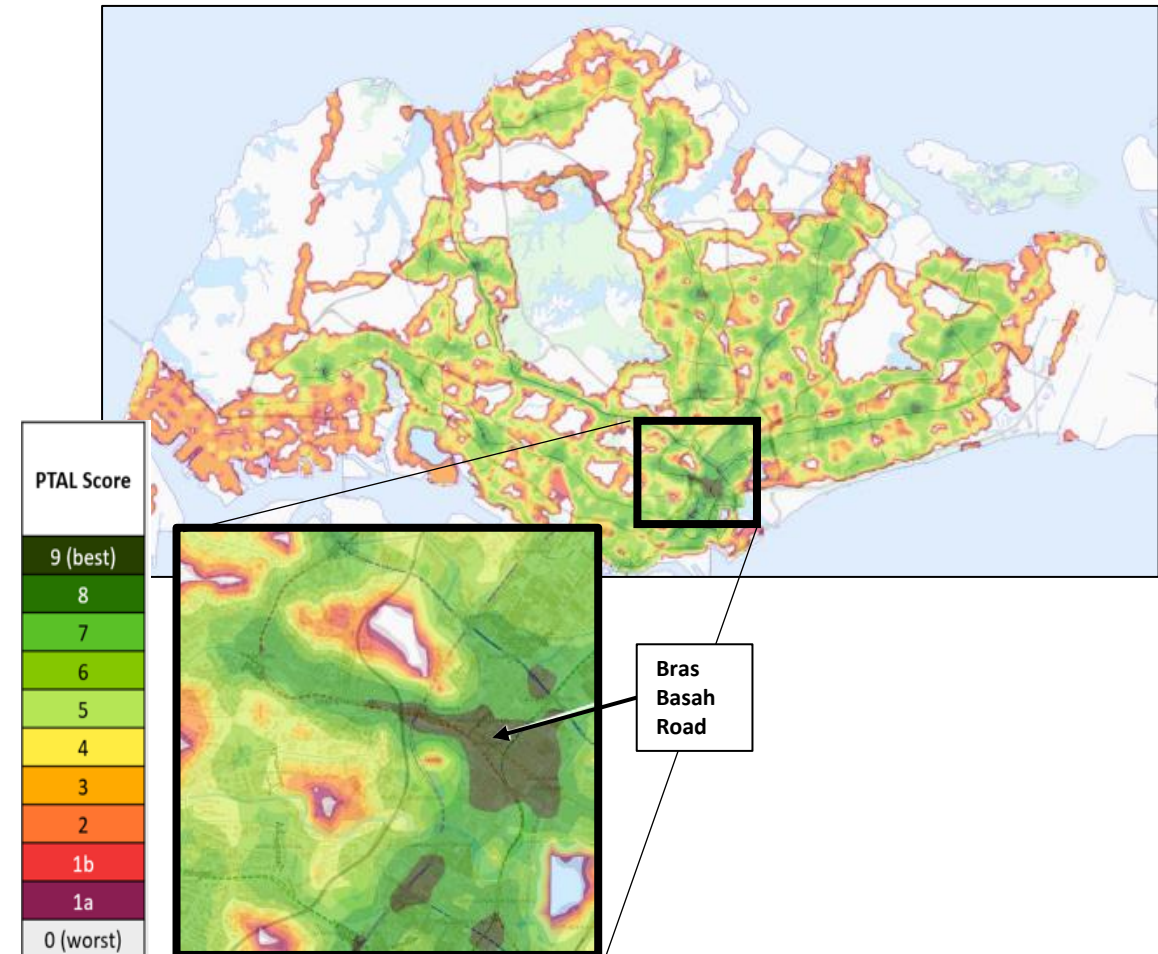
High car ownership areas



Should car ownership be discouraged in areas with **high** PTAL score and **high** car ownership? How?

Public transport connectivity for a location (PTAL)

- A composite score to measure the level of public transport connectivity for a location
- **Inputs:** PT services (choice), walking distance (proximity) and waiting time (frequency)
- **Output:** PTAL score (0-9); the higher the score, the better
- **Use:** Identify gaps, assess connectivity, develop policies/strategies/standards

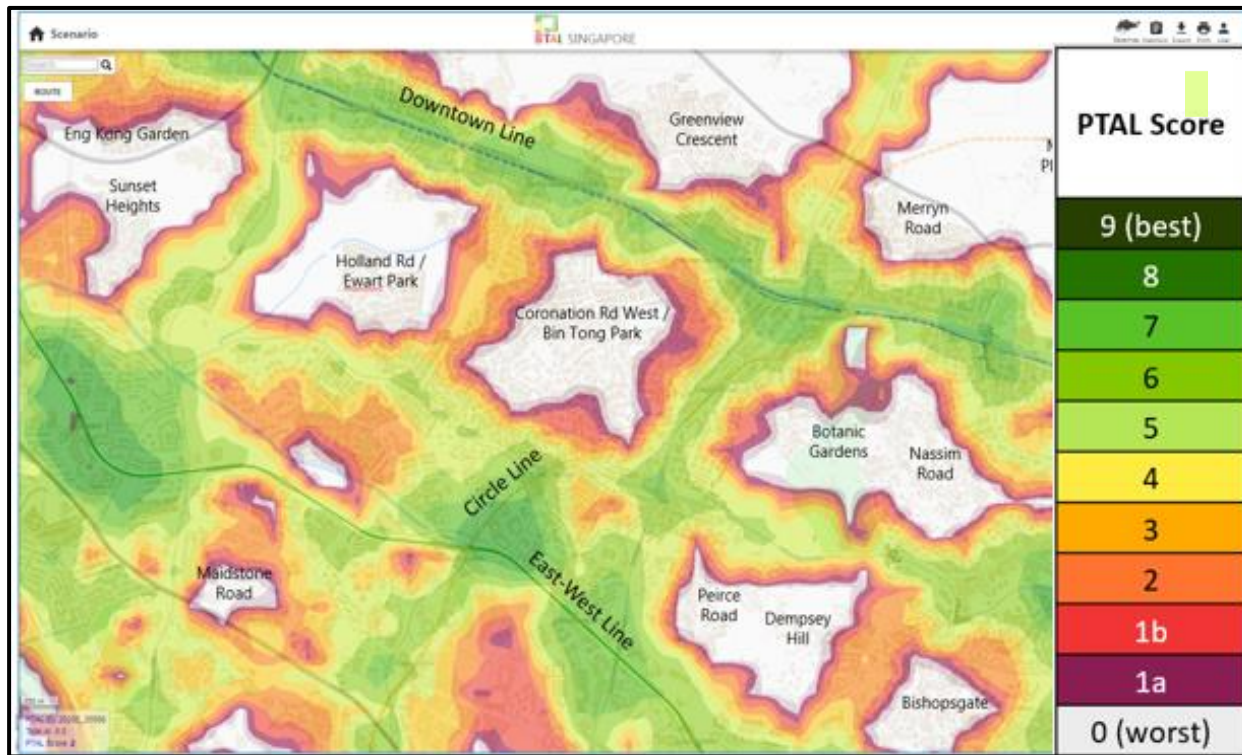


Source and image credits: LTA Singapore 2025

Examples of the use of PTAL

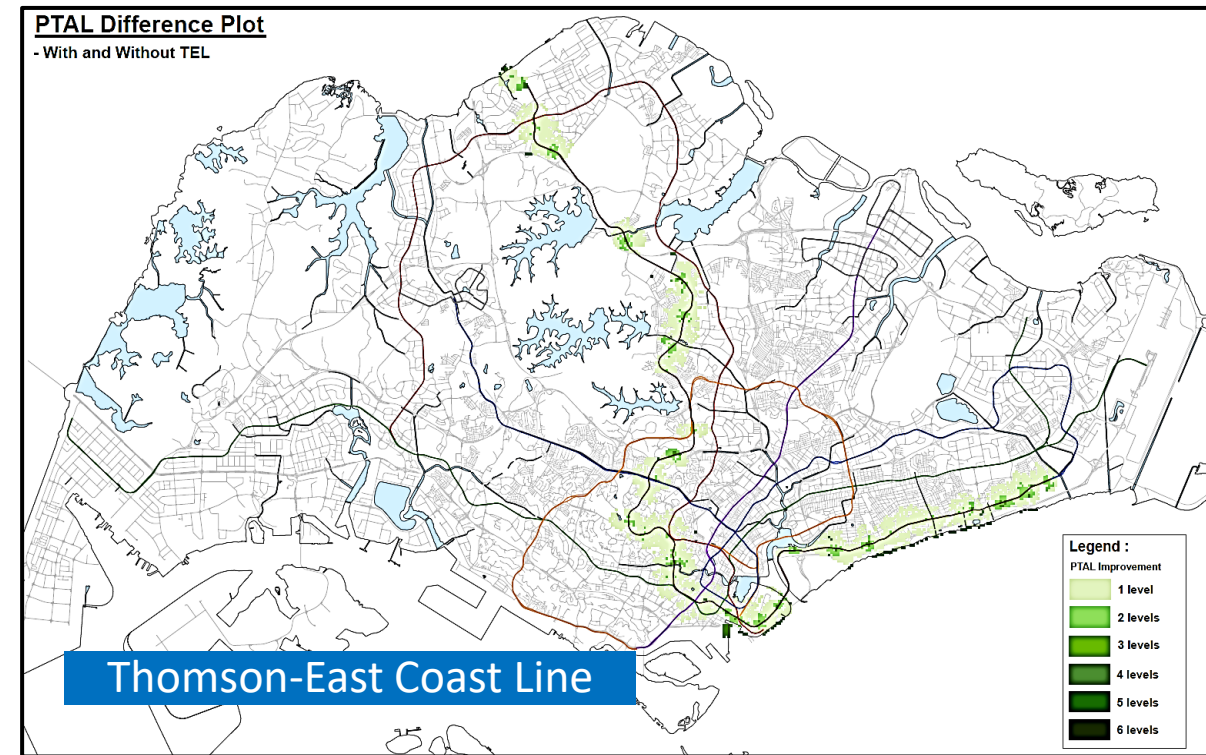
PT gap-finding

Helps identify suitable policy/scheme interventions for low or high PT accessibility areas



PTAL impact of new RTS lines

Helps understand PT accessibility implications of proposals as part of the transport appraisal



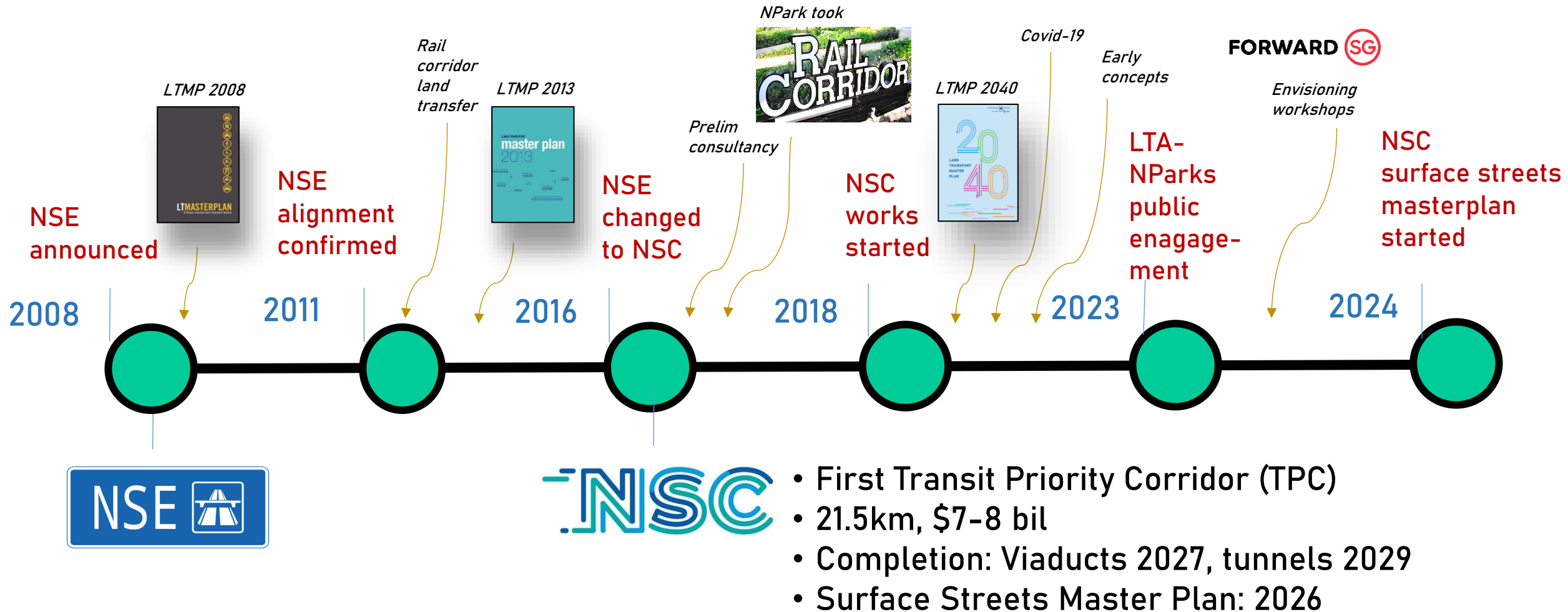
What are some case examples?

North-South Corridor (NSC)



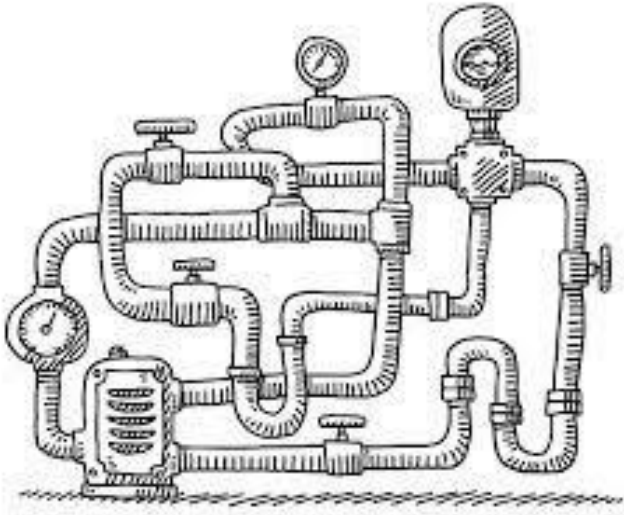
Image source: LTA
Singapore 2025

Transformative implementation of NSC



Source: Looi TS researched from various public sources, 2025

Re-imagine: From an expressway ...



The 11th expressway for high capacity,
high speed traffic movements



Source and image credit: LTA Singapore, 2025

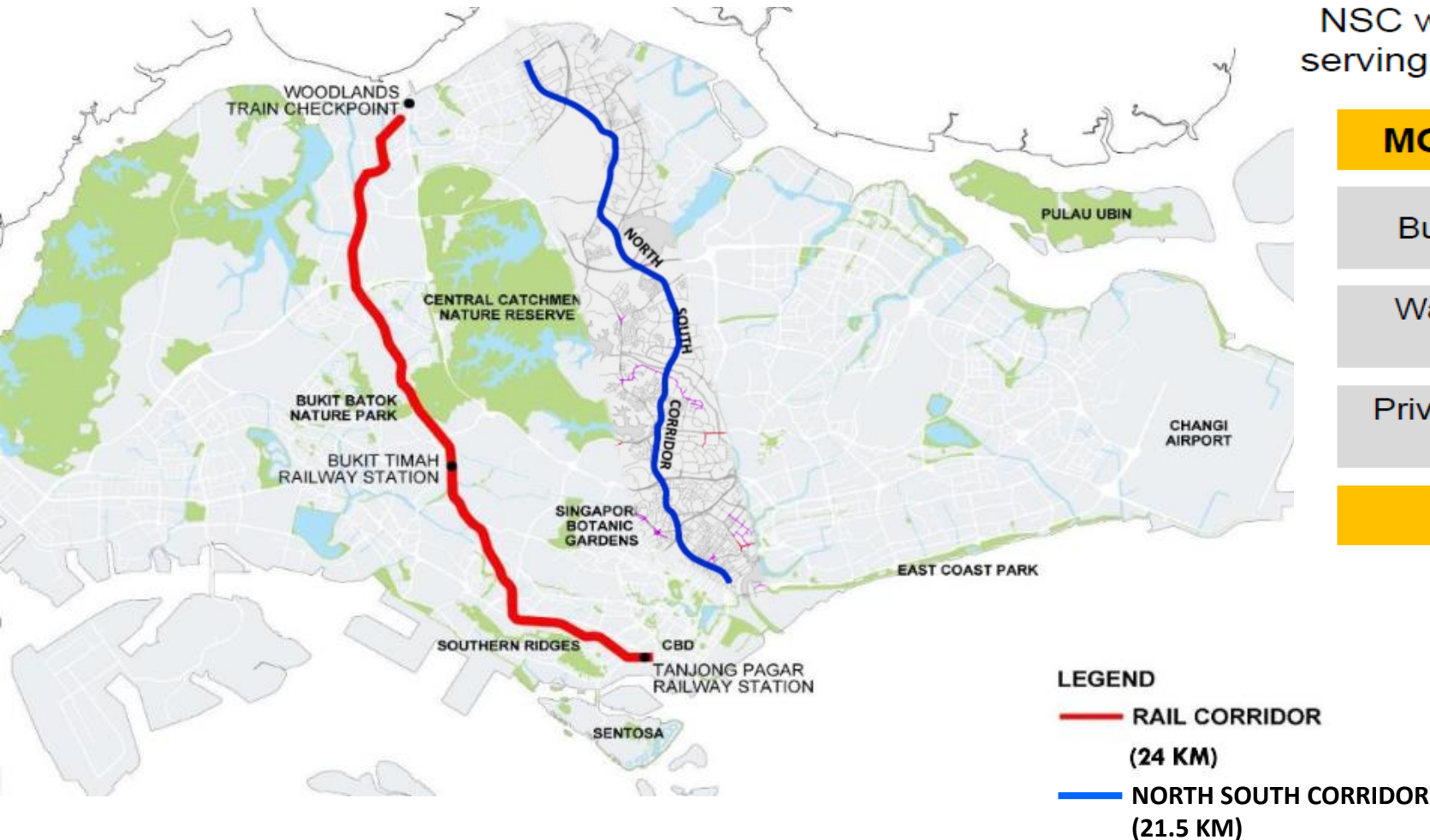
... to a delightful corridor



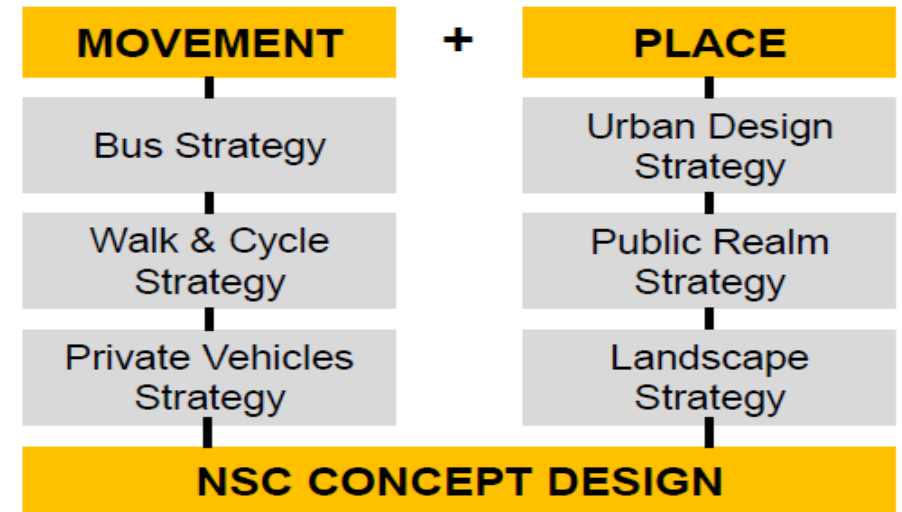
Source and image credit: LTA Singapore, 2024

NSC as a game-changer mega project

Shared vision of connecting people, places and possibilities



NSC was developed on the basis of a street serving a dual function – movement and place



NSC Surface Streets Master Plan

Completion: 2026

Source: LTA Singapore, 2025

Now: Singapore's life reserve – A place for all

4 distinct character segments, 20 public nodes



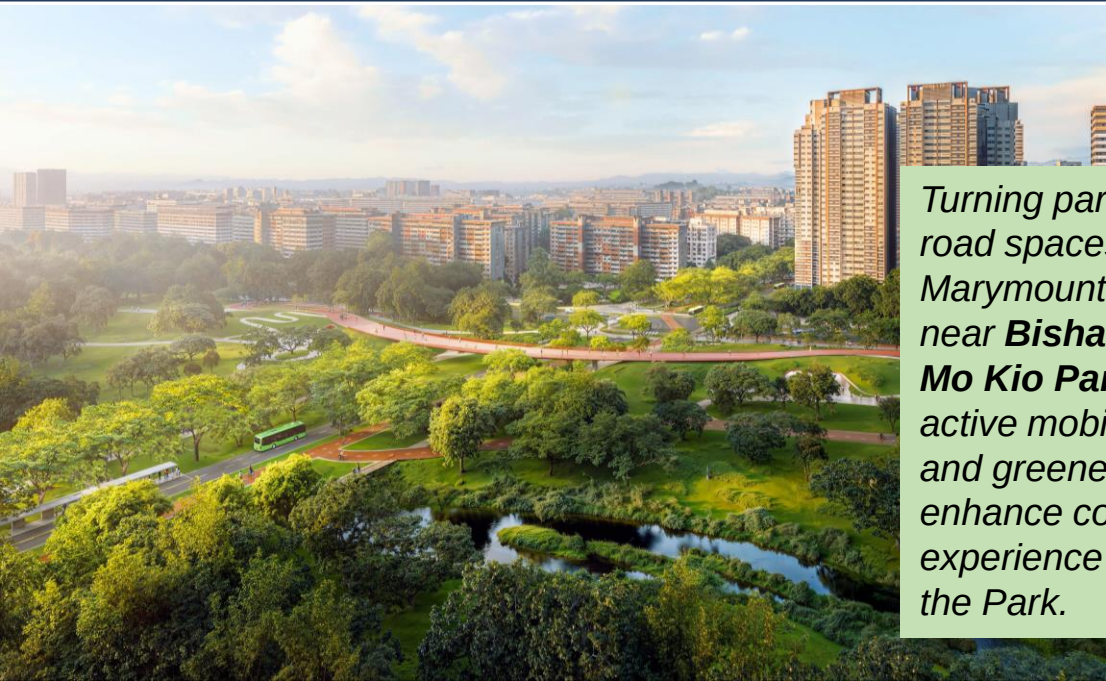
“...We are crafting a living, breathing space that will be a blueprint for future mobility and urban living, one that can inspire similar transformation in cities across the world.”

- Leonard Ng, Henning

Larsen



Prelim ideas of Ecological Loop segment at Ang Mo Kio-Bishan



*Turning part of the road spaces along Marymount Road near **Bishan-Ang Mo Kio Park** into active mobility uses and greenery to enhance commuting experience along the Park.*

Source & image credits: LTA Singapore, 2025



Prioritising public transport travel and reimagining surface streets for walking, cycling, public spaces and greenery as vehicular traffic is diverted into tunnel.



Lush planting with meandering paths to create a unique commuting experience.



Prelim ideas of Cultural– Heritage segment at Rochor-Ophia

Rochor Road Area –
Proposed public space
with enhanced walk-
cycle-ride experience to
connect the Little India,
Kampong Gelam and
Waterloo Street area.



Ophir Road (stretch
along Kampong Gelam)
– Transforming the road
into a public space for
events, activities, and to
connect two key
precincts - Kampong
Gelam and Bugis.



Source & image credits: LTA Singapore, 2025

Challenging & multi-disciplinary scope

Work-in-progress...



CONNECTIVITY PLAN



ENGAGEMENT PLAN



PLACEMAKING STRATEGY



GREENERY STRATEGY



DESIGN GUIDE

Research and innovation?

Urban heat island effect?



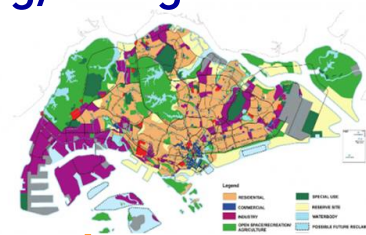
Impact on real estate?



Better health & wellness?



New planning/design model?

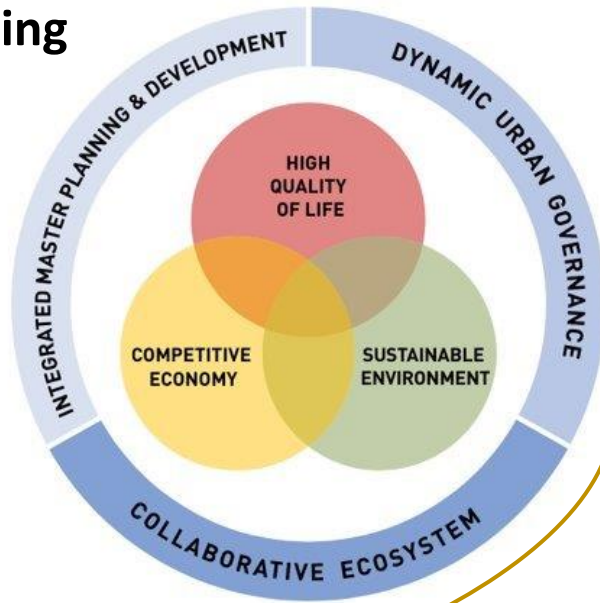


Source: LTA Singapore, 2025

Linking to urban systems guiding principles

Integrated Master Planning & Development

1. Think long term
2. Build for resilience
3. Seek “win-win”
4. Execute effectively
5. Innovate systemically



Dynamic Urban Governance

1. Lead with vision and pragmatism
2. Build a culture of integrity
3. Cultivate sound institutions
4. Engage and partner with the community
5. Work with markets

Collaborative Eco-system

1. Collective stewardship as a foundation for transformative collaborations
2. Forging strong international partnerships



Friendly streets for local communities



- **Friendly streets** to encourage gracious road-user behaviour, making residential neighbourhoods more inclusive, people-friendly and safe
- Engaged and co-created with local communities
- Part of **Town Mobility Plans**
- Piloted in 5 towns (Ang Mo Kio, Tampines, Toa Payoh, West Coast and Bt Batok West)
- 10 more coming up from 2025

10 upcoming Friendly Streets from 2025

Friendly Streets

Introducing
10 MORE FRIENDLY STREETS

THE LOCATIONS

1	Bedok North St 2	6	Jurong East St 24
2	Buangkok Crescent	7	Owen Rd, Cambridge Road, Kent Rd, Dorset Rd, Gloucester Rd
3	Bangkok Rd/Bukit Panjang Ring Rd/Pending Rd	8	Punggol Dr
4	Choa Chu Kang Ave 2	9	Admiralty Dr, Sembawang Dr/Way/Vista/Alley
5	Holland Ave/Dr, Holland Village Way, Lorong Liput	10	Havelock Rd

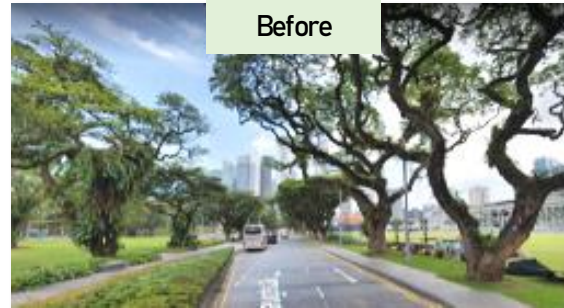
For artist's impressions and street maps, click [HERE](#).

Source: LTA Singapore, 2025

Road repurposing for better street spaces

- **Road repurposing** to dedicate more space to active mobility, public transport and community uses
- Engaged and co-created with local communities
- Part of **Town Mobility Plans**
- Target 60 projects by 2030

Civic District



Havelock Road



Tiong Bahru



Viaduct repurposing for active mobility



Source and image credits: ST 13 Nov 2022, LTA Singapore Nov 2022

- 1 km of 'to-be-defunct' MRT viaduct and associated spaces below it
- Instead of demolishing it, possible to repurpose it for active mobility or green corridor
- Feasibility study in-progress

Now, what's next?

Post-pandemic: Fundamentals remain sound

Actively shape new norms into shifts that reinforce long-term strategies

1

Integrated
planning



2

More
connections &
better PT



3

Liveable and
inclusive
community



4

Reduce reliance
on private
transport



5

Sustainable &
green land
transport



Challenges with no easy answers

Complex trade-offs in **choices**, **costs**, **climate change**, **community needs**



- Increasing **complexity** of land constraint issues
- **Changing mindsets** and making car-lite lifestyle a norm
- **Evolving trends** in work, live and play that shapes travel patterns
- Ensuring our transport system meet the **diverse population needs** and better support our aging and more vulnerable population
- Preparing for **technological shifts, disruptions** and **systems resilience**

Once over lightly

- Transport in a city is complex
- How do we solve it?
 - Planning
 - Public transport
 - Private transport
 - Value of data
- Case examples (NSC and others)
- Challenges with no easy answers



Some useful links

Liveability Framework

<https://www.clc.gov.sg/research-publications/publications/books/view/building-liveable-and-sustainable-cities--a-framework-for-the-future>

USS Transport

[https://www.clc.gov.sg/research-publications/publications/urban-systems-studies/view/transport-overcoming-constraints-sustaining-mobility-\(revised\)](https://www.clc.gov.sg/research-publications/publications/urban-systems-studies/view/transport-overcoming-constraints-sustaining-mobility-(revised))

LTA Website

<https://www.lta.gov.sg/content/ltagov/en.html>

LTMP 2040

https://www.lta.gov.sg/content/ltagov/en/who_we_are/our_work/land_transport_master_plan_2040.html

North South Corridor

https://www.lta.gov.sg/content/ltagov/en/newsroom/2024/8/news-releases/master_plan_consultant_appointed_for_nsc_surface_streets.html

Rail Corridor

<https://railcorridor.nparks.gov.sg/>

Friendly Streets

https://www.lta.gov.sg/content/ltagov/en/upcoming_projects/road_commuter_facilities/friendly-streets.html

Road Repurposing

https://www.lta.gov.sg/content/ltagov/en/upcoming_projects/road_commuter_facilities/road_repurposing.html

Tanah Merah MRT Viaduct

https://www.lta.gov.sg/content/ltagov/en/getting_around/public_transport/a_better_public_transport_experience/transforming-tanah-merah-mrt-viaduct-into-community-spaces.html

Jurong Regional Line (JRL)

https://www.lta.gov.sg/content/ltagov/en/upcoming_projects/rail_expansion/jurong_region_line.html

Thomson East Coast Line (TEL)

https://www.lta.gov.sg/content/ltagov/en/upcoming_projects/rail_expansion/thomson_east_coast_line.html

Cross-Island Line (CRL)

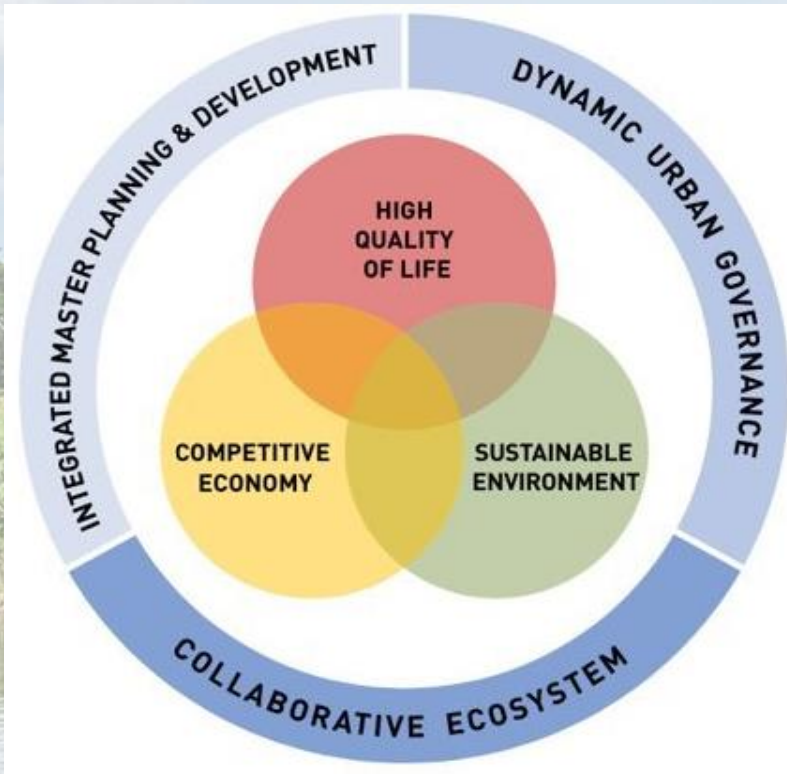
https://www.lta.gov.sg/content/ltagov/en/upcoming_projects/rail_expansion/cross_island_line.html

Cycling

<https://www.lta.gov.sg/content/ltagov/en/map/cycle.html>

https://www.lta.gov.sg/content/ltagov/en/getting_around/active_mobility/walking_cycling_infrastructure/cycling.html

Coming back to the two CDE 2501 take-aways



- **Understand** how the urban **systems approach** has helped Singapore achieve its liveability outcomes (*See the big picture*)
- **Appreciate** your **role** as future professionals to solve urban problems (*Offer your domain depths*)

Tips:

- Be **inquisitive**, ask questions and **discover** to learn.
- Make full use of your time in the course to **enjoy yourselves**.

Q&A

Awards for best projects

Best projects will be given an award!
More details will be released soon!



Important Information

1. Canvas and studio guide are essential materials for this course – please be familiarised with them
2. Tutorial/Studio Allocation:
 - Add/swap/appeals to refer to Canvas announcement for instructions
 - Further allocation will be done after Round 2 to ensure a diversity of discipline in each class. Your selected day and time will remain
 - **Teams and studio class rosters** will be finalised by **27 January 2025**. Do watch out for announcements on Canvas 😊